

FF FaceFind

Find Your Photos. Instantly:

AI-powered face recognition for event photography.
Guests scan a QR code, take a selfie — find every photo of themselves in seconds.

CS 691

Team 7

Deliverable 3

TEAM

Member Roles & Responsibilities

Jash Berawala

Full Stack Lead

- Backend API (Node.js/Express)
- GitHub Management
- ML Integration
- API Development & Testing

Jay Shah

Frontend Developer

- React.js Application
- Admin Dashboard Interface
- Guest Photo Gallery
- Responsive Design & State Mgmt

Niraj Patil

ML Engineer

- Face Detection & Recognition
- Python Microservice (FastAPI)
- Face Clustering (DBSCAN)
- Model Performance Optimization

Chetan Patel

Database Admin / BA

- AWS S3 Integration
- Database Architecture (MongoDB)
- Docker Containerization
- Security & QR Code System

Improvements made from Professor's Feedback



Improvements



Organized all slides in their appropriate sections.

PROBLEM

Event Photos Are Impossible to Find.



Thousands of Photos

Photographers upload 1,000–5,000+ images per event.



No Way to Search

Guests have no tool to find photos of themselves.



Hours Wasted

Manual searching kills the joy of the moment.

“

I spent 2 hours scrolling through 3,000 wedding photos just to find 12 pictures of myself.

— Event Attendee

94%

of guests never find
all their photos

SOLUTION

FaceFind. Scan. Snap. See Yourself.

01



Photographer Uploads

Creates an event, uploads all photos. AI processes and indexes every face automatically.

02



Guest Scans QR

Attendees scan the unique QR code at the venue to access the guest portal instantly.

03



Takes a Selfie

Upload or take a selfie directly in the browser — no app download required.

04



Instant Match

AI matches your face against all event photos and returns your results in under 5 seconds.

⚡ Results delivered in under 5 seconds · No app download required · Works on any device

Built for Photographers, Loved by Guests.



AI Face Indexing

Every uploaded photo is automatically processed and indexed by face embedding.



< 5 Second Match

Guests receive matched results in under 5 seconds using vector similarity search.



No App Required

Fully browser-based. Guests just scan a QR code — nothing to install.



Photographer Auth

Secure JWT-based login. Each photographer sees only their own events.



Live Dashboard

Real-time processing updates via WebSockets. See photos processed as they upload.



QR Code Sharing

Auto-generated QR codes per event. Share with guests in seconds.

Our Commitment to Each Other.

1 Communication

- Primary platform: Microsoft Teams / Google Meet
- Respond to messages within 12 hours
- Blockers & delays communicated promptly and transparently

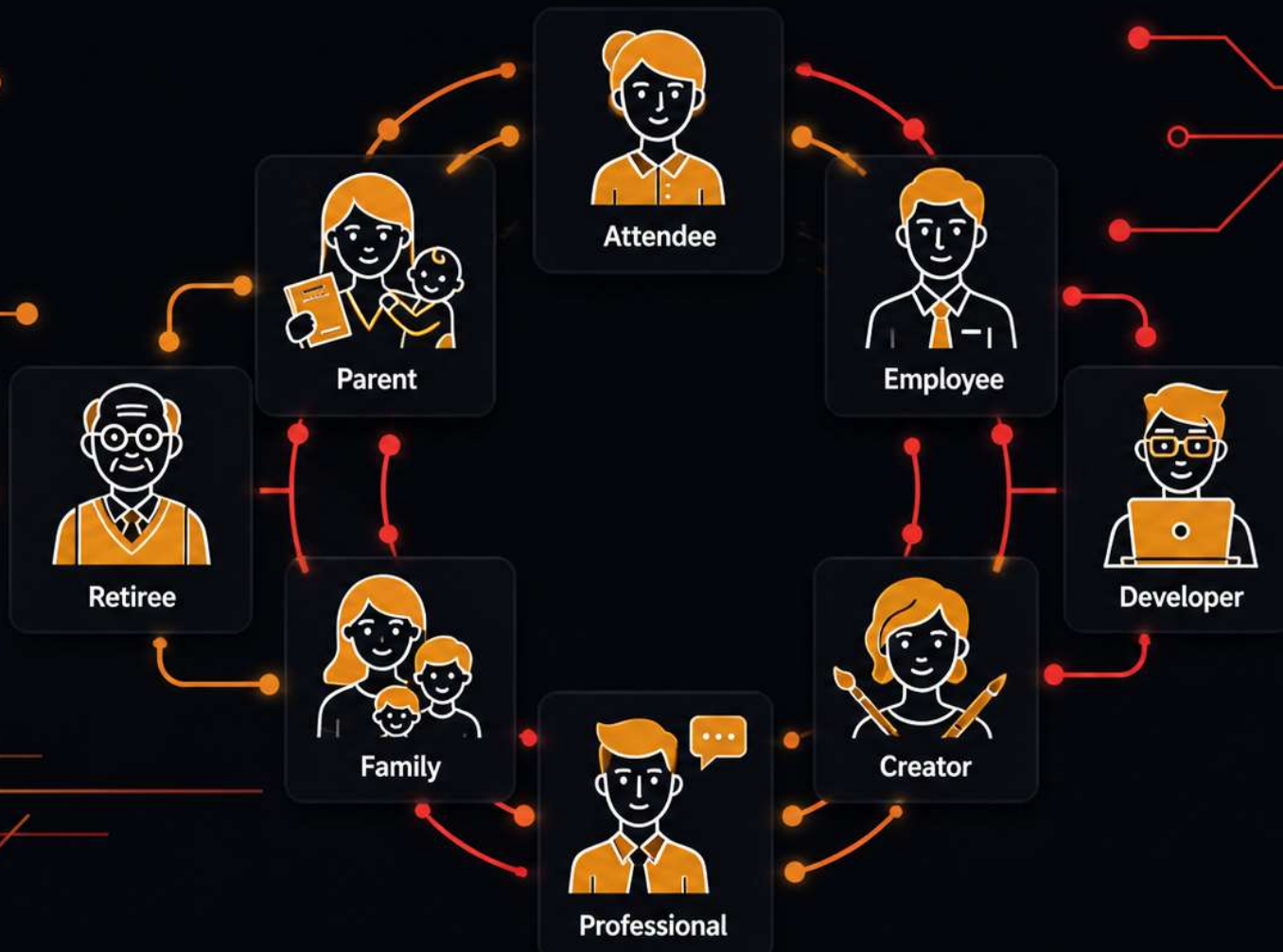
2 Meetings

- Attend all sprint ceremonies: Planning, Backlog Grooming, Reviews
- Arrive prepared with updates and blockers
- Retrospectives conducted with candor and respect

3 Work Expectations

- Complete assigned tasks before the deadline
- Submit work for peer review before merging to main
- Communicate blockers early so the team can adjust

Personas





DEMOGRAPHICS



Name: Alex Reynolds



Age: 31



Location: Denver, CO



Business: Moments by Alex Photo



PROFILE

- Full-time Event Photographer
- 8 years experience
- Shoots 3-4 events/week
- 800-1200 photos per event
- Specializes in weddings & corporate events



GOALS

- Reduce photo delivery time from hours to minutes
- Improve client satisfaction with faster turnaround
- Focus more on photography and client acquisition
- Increase bookings through referrals
- Build brand via photo tags



FRUSTRATION

- Manual folder creation and link sharing takes 5-6 hours per event
- Constant client requests: "When will photos be ready?"
- No easy way to provide partial access during events
- Link sharing management is cumbersome
- Lost referral opportunities from untagged photos



HOW FACEFIND HELPS

- Creates event in 30 seconds
- AI handles photo distribution
- Guests self-serve via QR code
- Real-time processing during breaks
- Automatic tagging for referrals
- Frees 5+ hours per event



QUOTE



I spent 6 hours last night just organizing and sending photos from Saturday's wedding. That's time I could have been shooting, marketing, or actually living my life. If guests could just scan a QR code and instantly get their photos, I'd be a hero."



— Alex Reynolds, Professional Photographer



DEMOGRAPHICS



Name : **Marcus Thompson**



Age : 28



Location : Atlanta, GA



Occupation : Marketing Manager



Education : MBA, Emory University



PROFILE

- Marketing Manager with 5+ years experience
- Attends 8-10 weddings annually
- Active on Instagram & LinkedIn
- Posts within 24 hours for max engagement
- Values professional quality photos



FRUSTRATION

- Receives photo links 3-4 weeks after events.
- Must scroll through 800+ photos to find himself
- Misses social media engagement window.
- Phone photos are poor quality.
- Takes photos during events instead of enjoying



GOALS

- Get photos immediately while memories are fresh
- Easily find photos of himself without scrolling
- Share to social media during the event
- Enjoy events without phone distraction
- Build social media presence



HOW FACEFIND HELPS

- Scan QR code at event entrance
- Upload one selfie
- AI finds ALL photos of Marcus
- Instant access to professional photos
- Share during event when engagement peaks
- Selfie deleted immediately after matching



QUOTE



"I just got the photo link from my cousin's wedding THREE WEEKS after the event. The moment has passed, everyone's moved on. Scrolling through 900 photos to find the 3 where I look decent is exhausting. Why can't someone just send me MY photos?"



— **Marcus Thompson**, Wedding Guest



DEMOGRAPHICS



Name : **Jordan Chen**



Age : 40



Location : Chicago, IL



Company : Global Tech Solutions



Position : Senior Events Manager



PROFILE

- Senior Events Manager with 12 years experience
- Organizes 15-20 corporate events annually
- Manages \$3M annual event budget
- Average event: 100-500 attendees
- Reports to C-suite leadership
- Data-driven decision maker



FRUSTRATION

- No way to track photo engagement or ROI.
- Privacy concerns with employee photos.
- Can't prove event value with hard data.
- Post-event follow-up has low response rates.
- Budget justification difficult without metrics.
- Leadership asks: "What did we get for that investment?"



GOALS

- Quick, secure photo delivery to employees
- Track engagement and measure ROI
- Control access with email whitelisting
- Generate reports for leadership
- Ensure data privacy and compliance
- Demonstrate event value beyond attendance



HOW FACEFIND HELPS

- Private events with email whitelist
- Detailed analytics dashboard
- Exportable reports for leadership
- Automatic session expiration
- Corporate branding integration
- GDPR/CCPA compliant
- 73% engagement in 48 hours



QUOTE

"Last year's leadership summit cost \$250,000. When the CFO asked what we got for it, all I had was attendance numbers and some nice photos. Now I can show exactly how many people engaged, shared, and downloaded. That's how I justify next year's budget."

— **Jordan Chen**, Senior Events Manager



Minimum Viable Product.

✓ IN THE MVP

- ✓ Photographer register & login (JWT auth)
- ✓ Create & manage events with QR codes
- ✓ Upload photos → AI face indexing
- ✓ Guest selfie → instant photo match
- ✓ Real-time processing dashboard (WebSocket)
- ✓ Event catalog & browsing

🚀 FUTURE ROADMAP

- Payment integration for photo purchases
- AI-generated captions & watermarking
- Multi-photographer collaboration
- Mobile app (iOS / Android)
- Advanced analytics & reporting
- Social media auto-publish

Full-Stack Technology Choices.

Frontend

- React.js
- Redux Toolkit
- Tailwind CSS

Backend

- Node.js + Express
- MongoDB + Redis
- JWT + Socket.io

Machine Learning

- Python + PyTorch
- FastAPI
- InsightFace + OpenCV

Cloud & DevOps

- AWS S3 + CloudFront
- Docker Containers
- GitHub Actions CI/CD

ML / AI Algorithms Used.



InsightFace / ArcFace

Deep metric learning for face embedding generation. Produces 512-dim vectors for each face.



DBSCAN Clustering

Density-based spatial clustering to group similar faces. No predefined cluster count required.



Cosine Similarity Search

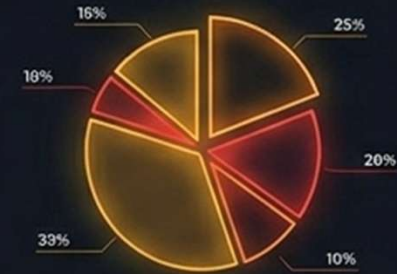
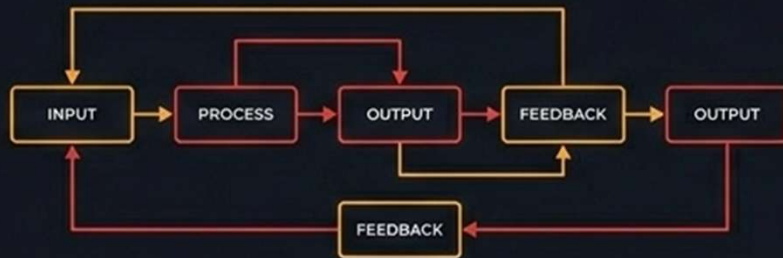
Vector similarity matching between guest selfie embedding and event photo embeddings.



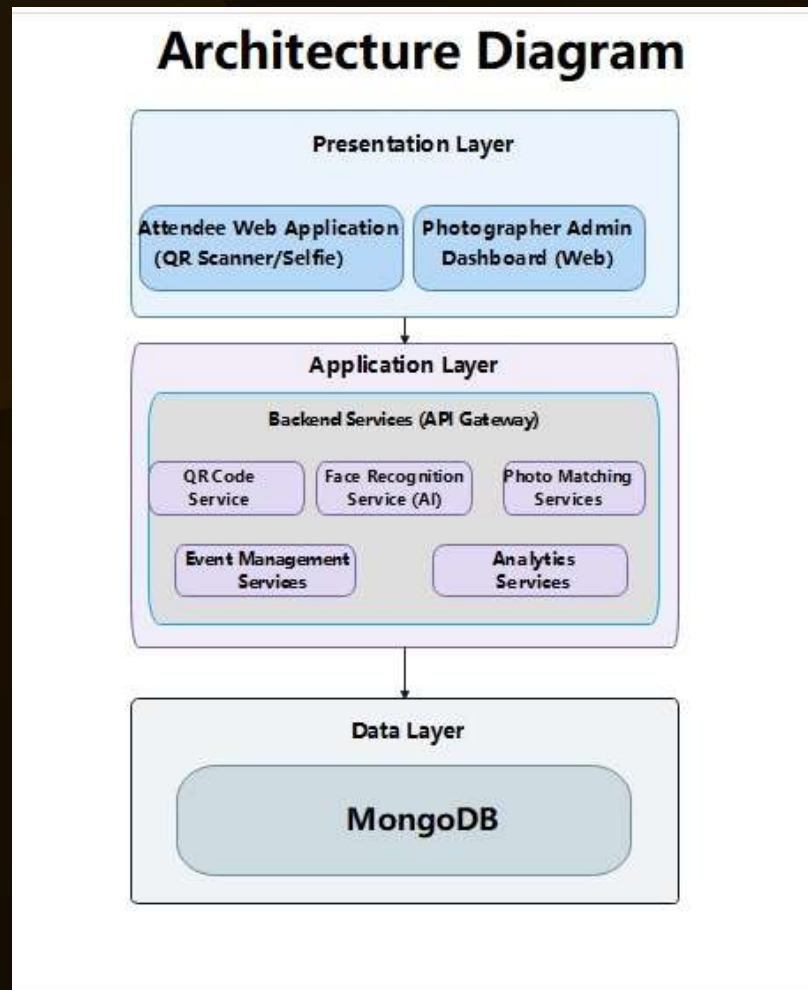
Real-time Processing

Async FastAPI pipeline with WebSocket notifications for live upload progress.

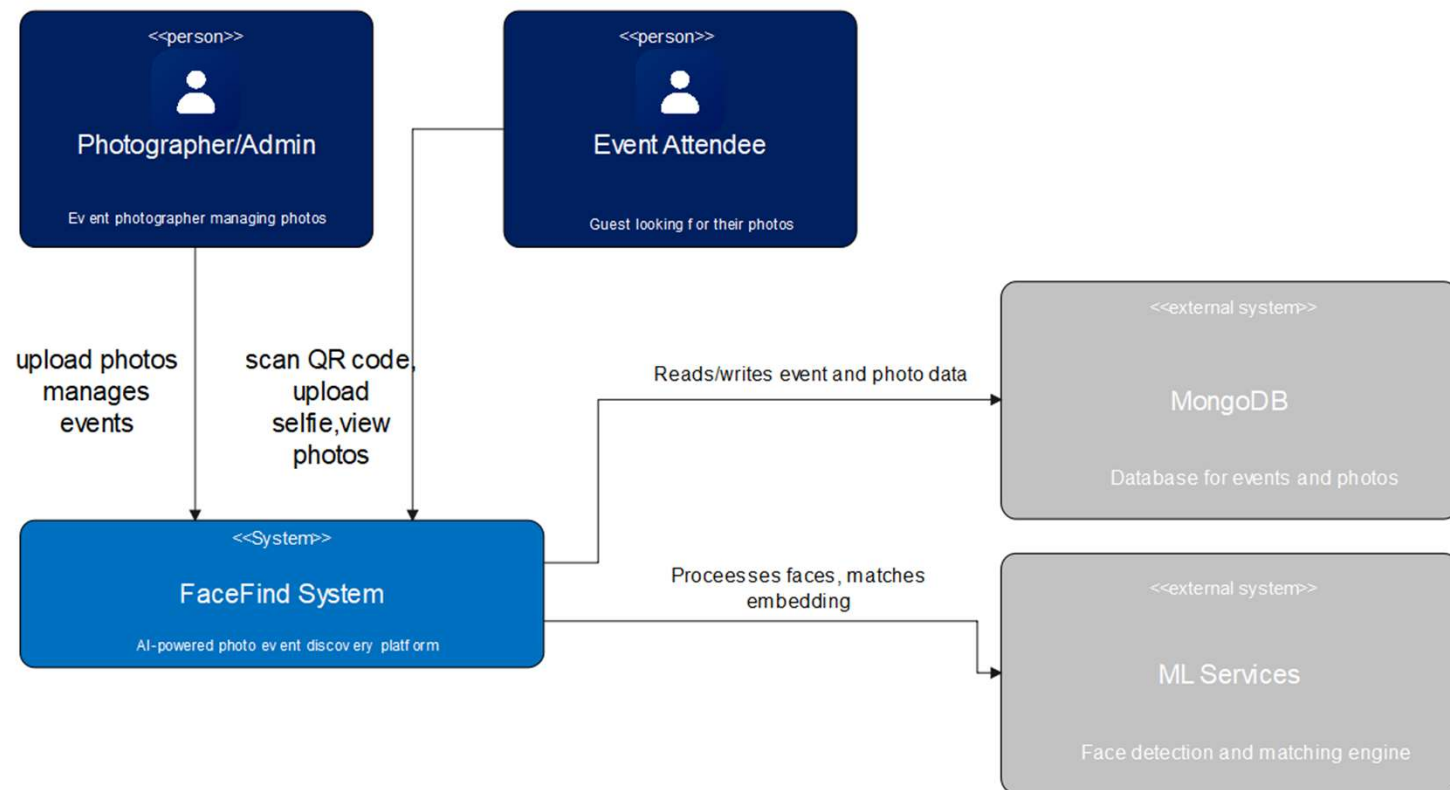
DIAGRAMS



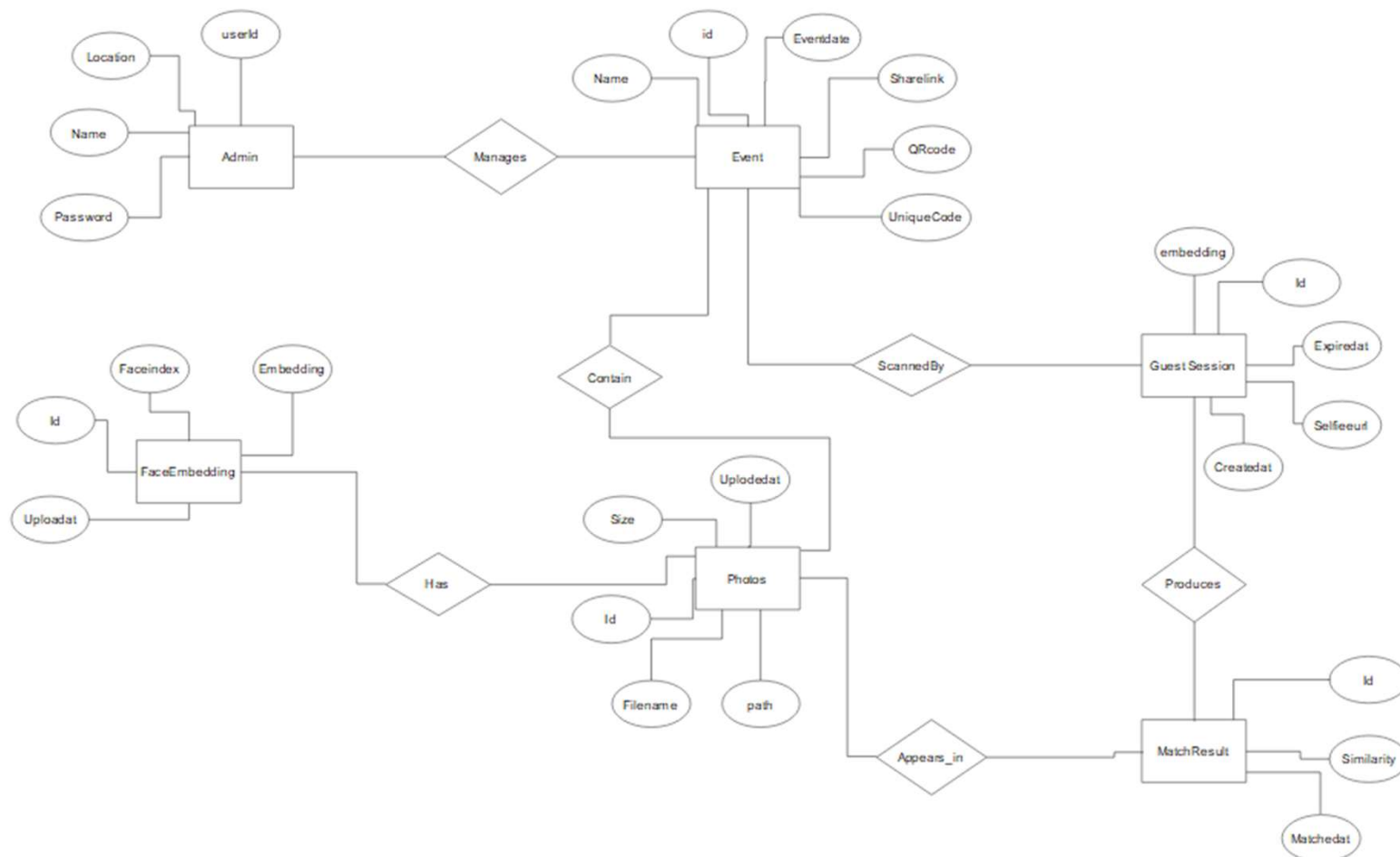
Architecture Diagram



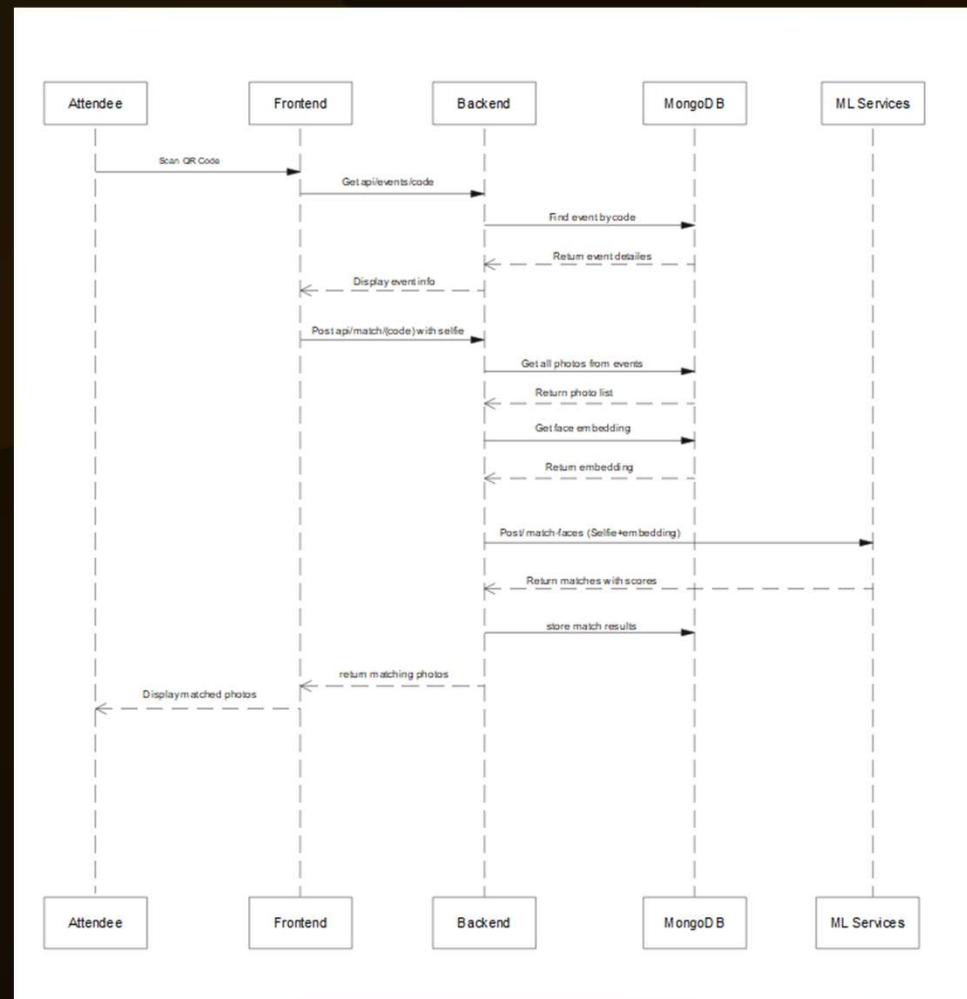
Context Diagram



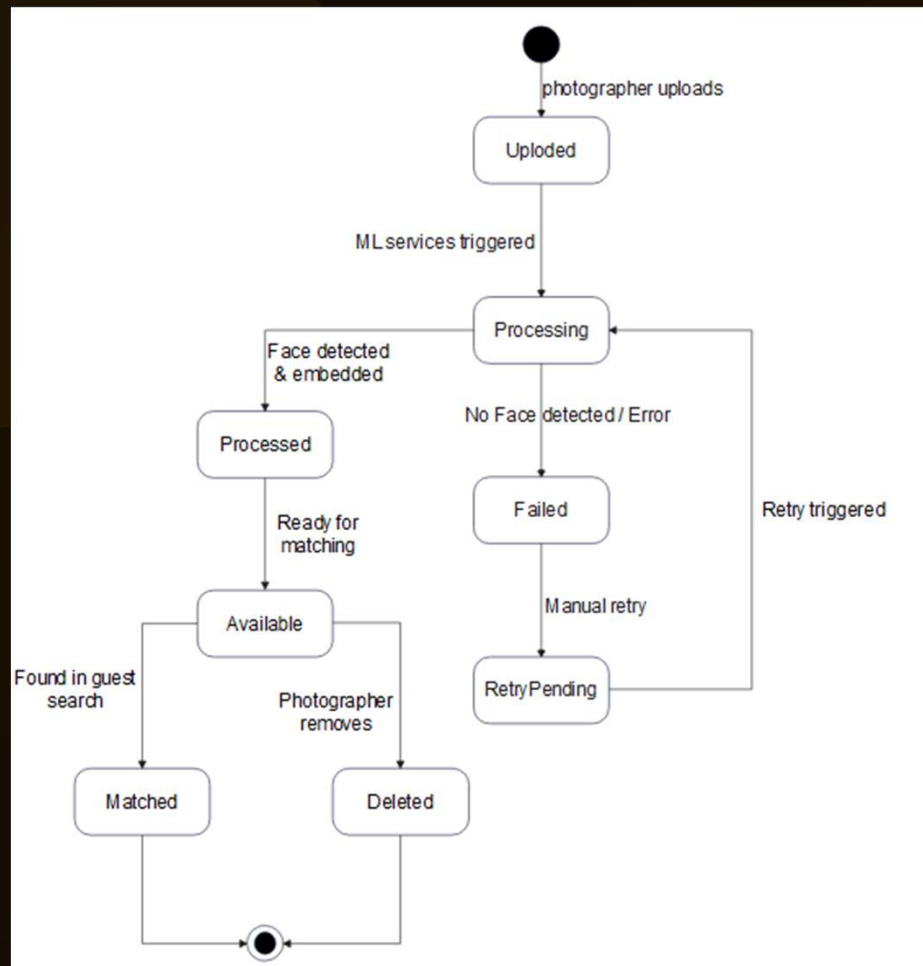
ER Diagram



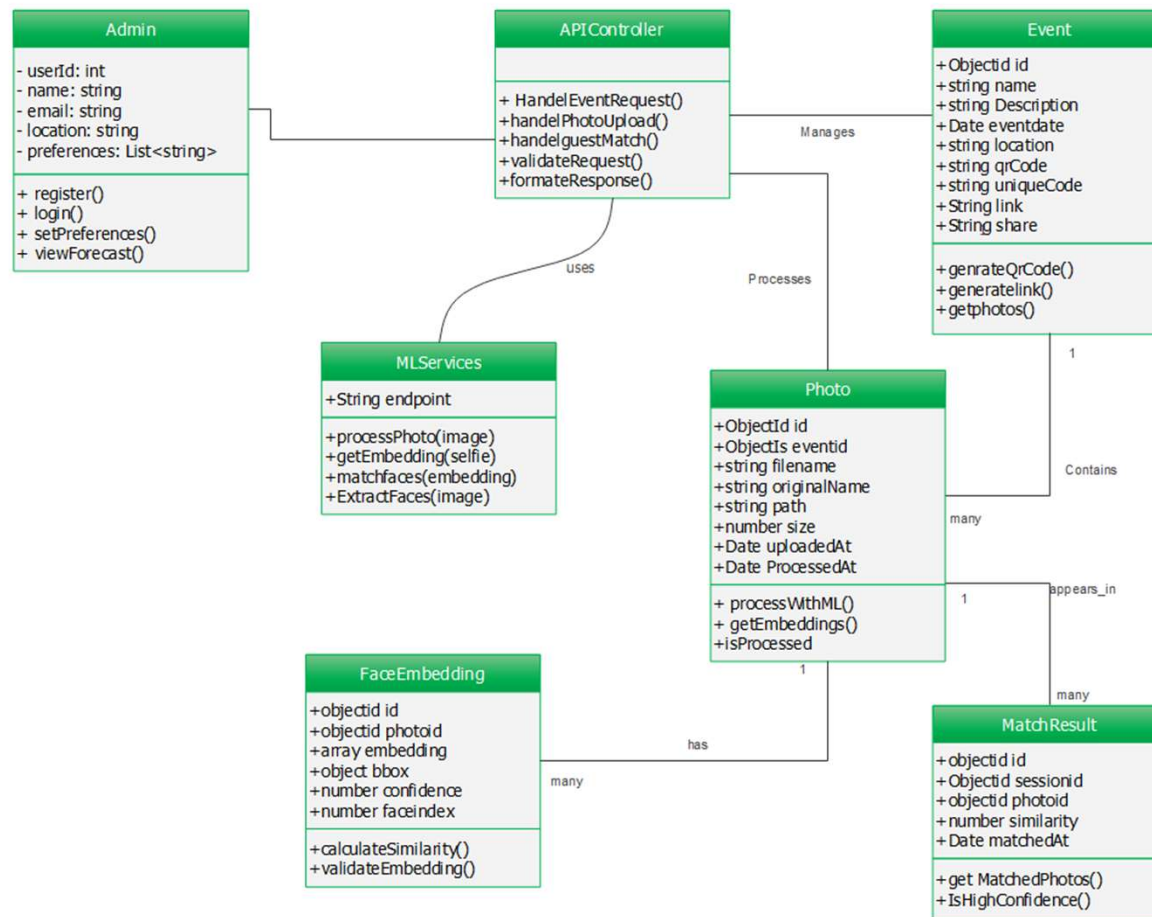
Sequence Diagram



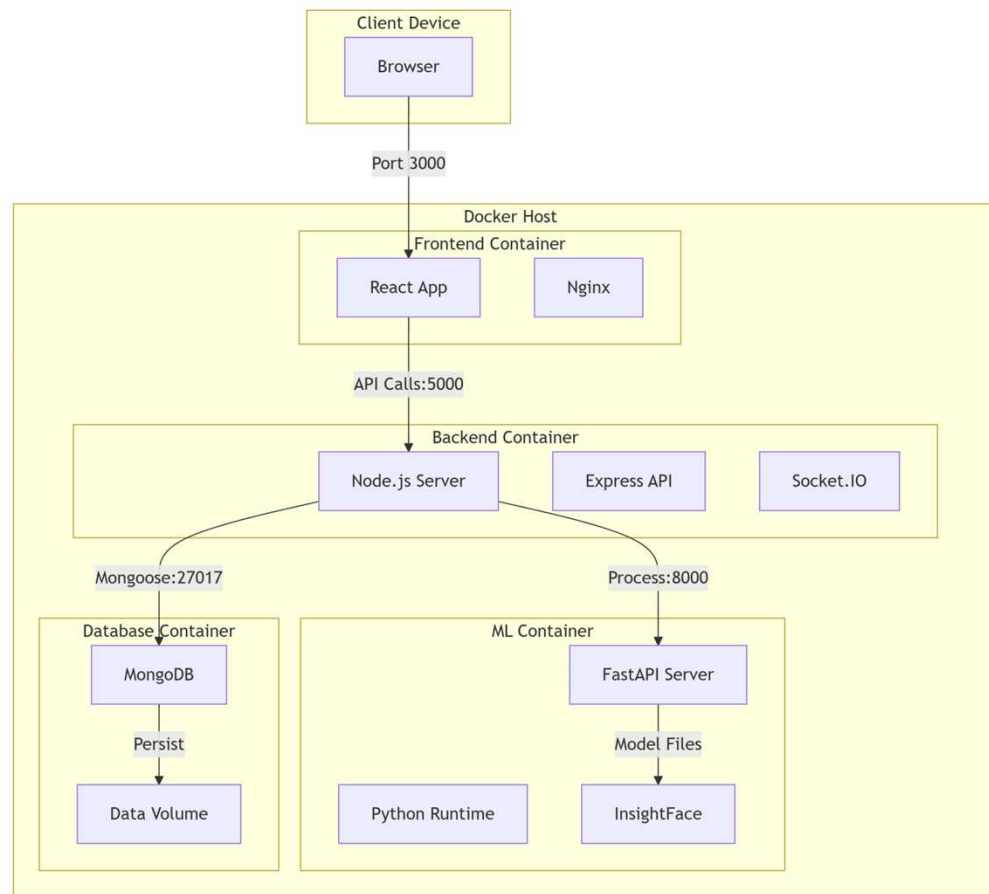
State Diagram



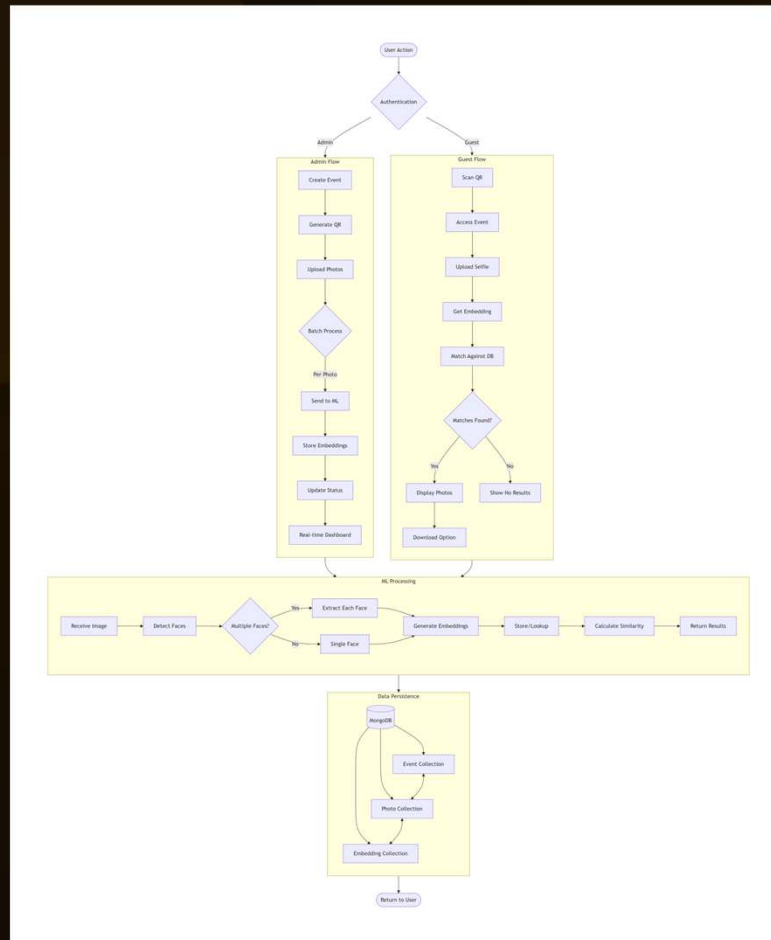
Class Diagram



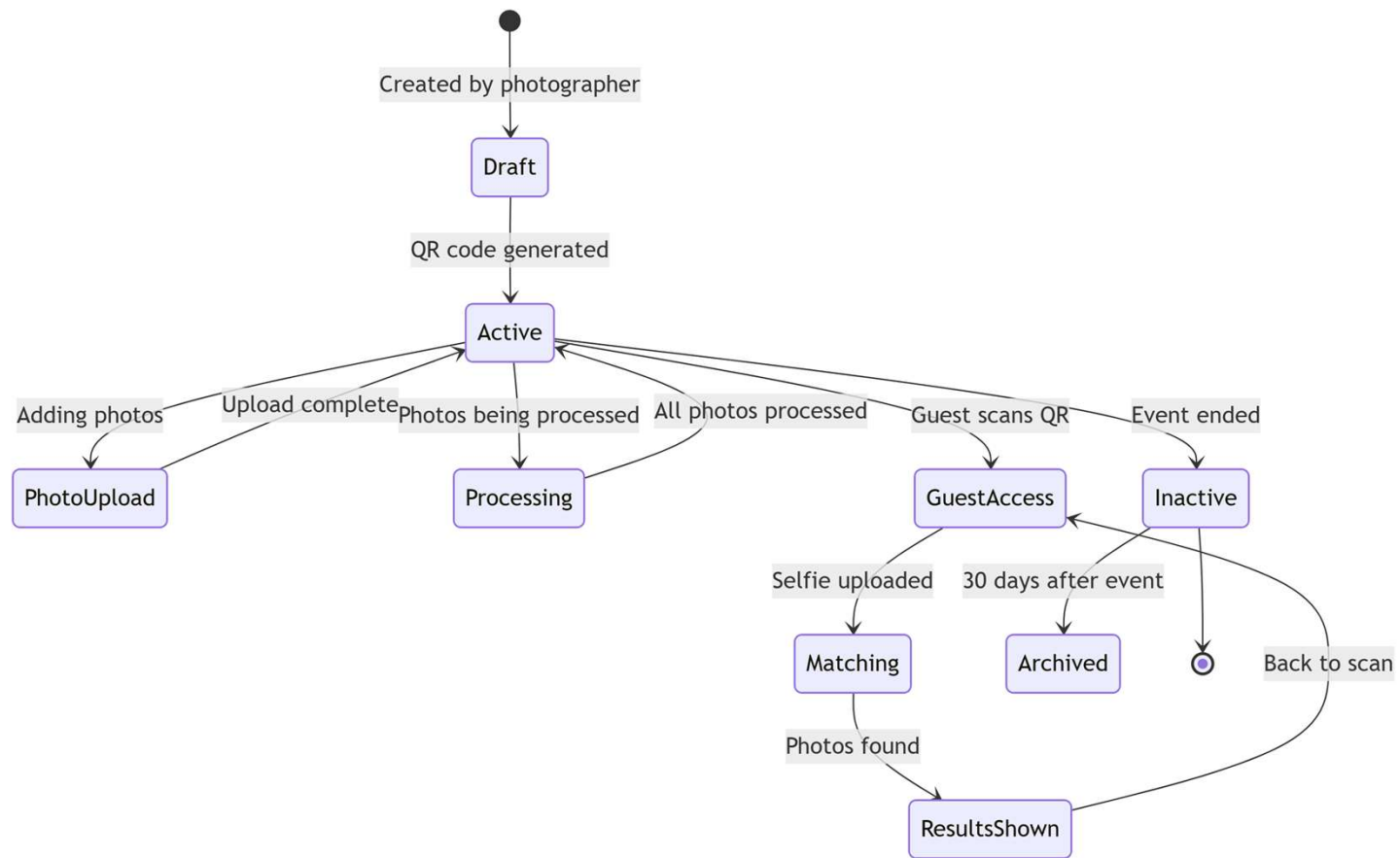
Deployment Diagram



Data Flow Architecture Diagram



State Diagram



User Stories — Summary

#	Epic	User Story	Pts	Status	Place
US-01	Account Management	As a photographer , I want to register an account with my name, email, studio, and password	3	✓ Done	Login/Register
US-02	Account Management	As a photographer , I want to log in with my email and password	2	✓ Done	Login
US-03	Account Management	As a signed-in photographer , I want to log out of my account	1	✓ Done	Logout
US-04	Account Management	As a photographer , I want to recover my forgotten password via email	3	✓ Done	Login
US-05	Account Management	As a photographer , I want to update my profile information (name, studio, email)	2	✓ Done	Profile
US-06	Account Management	As a photographer , I want to have my account and all data permanently deleted	3	✓ Done	Profile
US-07	Event Management	As a photographer , I want to create a new event with name, date, location, and description	5	✓ Done	Dashboard
US-08	Event Management	As a photographer , I want to view all my events in a list	2	✓ Done	Dashboard

US-11	Event Management	As a photographer , I want to generate and download a QR code for each event	3	✓ Done	Dashboard
US-12	Photo Upload	As a photographer , I want to upload up to 500 photos at a time to an event	8	✓ Done	Upload
US-13	Photo Upload	As a photographer , I want to see AI processing progress for each upload batch	5	✓ Done	Upload
US-14	Photo Upload	As a photographer , I want to drag and drop photos into the upload zone	3	✓ Done	Upload
US-15	Photo Upload	As a photographer , I want to delete individual photos from an event	3	✓ Done	Upload
US-16	AI Face Matching	As a guest , I want to upload a selfie at an event	13	✓ Done	Selfie Search
US-17	AI Face Matching	As a guest , I want to see all matched photos in a grid sorted by confidence	5	✓ Done	Results
US-18	AI Face Matching	As a guest , I want to download individual matched photos or all at once	5	✓ Done	Results
US-19	AI Face Matching	As a guest , I want my selfie to be automatically deleted after matching	5	✓ Done	Selfie Search
US-20	Event Catalog	As a visitor , I want to browse all public events on the catalog page	3	✓ Done	Catalog
US-21	Event Catalog	As a visitor , I want to search for an event by name or location	2	✓ Done	Catalog / Search
US-22	Event Catalog	As a visitor , I want to filter events by status and category	2	✓ Done	Catalog / Filter
US-23	Event Catalog	As a visitor , I want to see event details on each card (date, location, photo count)	1	✓ Done	Catalog

#	Epic	User Story	Pts	Status	Place
US-24	Dashboard & Analytics	As a photographer , I want to see key stats on my dashboard (events, photos, searches, active events)	3	✓ Done	Dashboard
US-25	Dashboard & Analytics	As a photographer , I want to see a recent activity feed	3	✓ Done	Dashboard
US-26	Dashboard & Analytics	As a photographer , I want to see per-event analytics (views, searches, match rate, downloads)	5	✓ Done	Analytics
US-27	Dashboard & Analytics	As a photographer , I want to export event analytics as a CSV	3	✓ Done	Analytics
US-28	Privacy & Security	As a guest , I want to be informed about how my selfie is used and deleted	3	✓ Done	Selfie Search
US-29	Privacy & Security	As a photographer , I want all API endpoints to require JWT authentication	5	✓ Done	API / Backend
US-30	Privacy & Security	As a photographer , I want to make an event private with an email whitelist	3	✓ Done	Dashboard
US-31	Privacy & Security	As a guest , I want to request deletion of all photos containing my face	3	✓ Done	Profile / Settings
US-32	Privacy & Security	As a photographer , I want to receive email confirmations for key account events	2	✓ Done	Profile

Account Management

US-01 Account Management · 3 pts · Done · Login/Register

As a	photographer,
I want to	register an account with my name, email, studio, and password
So that	I can keep my account private and log in securely.
Acceptance Criteria	→ Given I fill in all required fields and click "Create Account", then my account is created and I am redirected to the dashboard. → Given I enter an email that is already registered, then I see "Email already in use." → Given I enter passwords that do not match, then I see "Passwords do not match."

US-02 Account Management · 2 pts · Done · Login

As a	photographer,
I want to	log in with my email and password
So that	I can access my events and dashboard.
Acceptance Criteria	→ Given I enter valid credentials, then I am redirected to the photographer dashboard. → Given I enter incorrect credentials, then I see "Invalid email or password." → Given I check "Remember me", when I close and reopen the browser, then I remain logged in.

Event Management

US-07 Event Management · 5 pts · Done · Dashboard

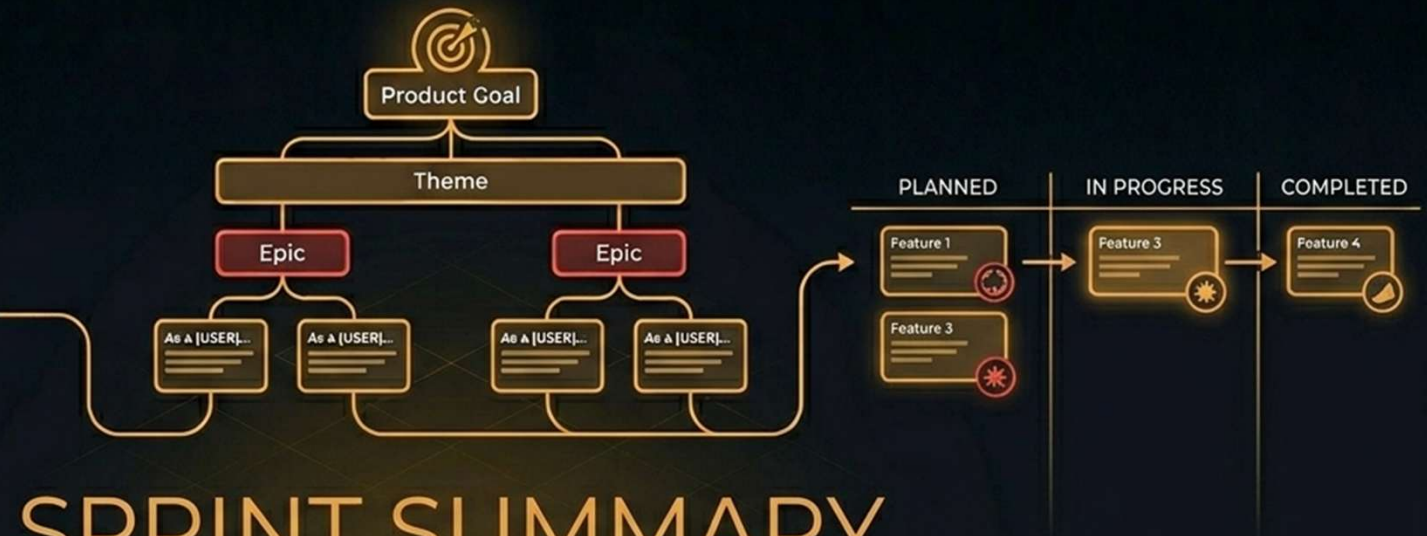
As a	photographer,
I want to	create a new event with name, date, location, and description
So that	I can organise my photo uploads by event.
Acceptance Criteria	→ Given I fill in event name and date and click "Create Event", then the event appears in my dashboard list. → Given I omit the event name, then I see "Event name is required." → Given the event is created, then a unique QR code and event code are automatically generated.

US-08 Event Management · 2 pts · Done · Dashboard

As a	photographer,
I want to	view all my events in a list
So that	I can manage and monitor them from one place.
Acceptance Criteria	→ Given I navigate to my dashboard, then I see all events sorted by most recently created. → Given I have no events, then I see "No events yet — create your first one!"



SPRINT PERFORMANCE



SPRINT SUMMARY

Backlog Refinement Area

KEY ACHIEVEMENTS



KEY BLOCKS



Deliverables 1

☐ Capstone Sprint 2	Add dates (10 work items)	0 0 49	Start sprint	...
☒ FF-17	Sprint 2 backlock investigation, and t...	DONE ▾	3	NP
☒ FF-18	Part 1: Preparing presentables for th...	DONE ▾	3	JB
☒ FF-20	jira work	DONE ▾	8	PC
☒ FF-21	Preparing Personas	DONE ▾	5	JB
☒ FF-22	Preparing Charts	DONE ▾	5	NP
☐ FF-24	Frontend Development – Landing Pa...	DONE ▾	8	= JS
☐ FF-25	Implement Login Authentication API ...	DONE ▾	8	= JS
☐ FF-28	Part 2: Preparing presentables for th...	DONE ▾	3	= JB
☐ FF-27	Investigation for next sprint doables	DONE ▾	3	= PC
☐ FF-26	Metrics	DONE ▾	3	= NP

Deliverables 1

Deliverables 1 Velocity Metrics.

49

Committed

story points

49

Completed

story points

0

Spillover

story points

49

Velocity

pts / sprint

Sprint Velocity — Story Points Completed

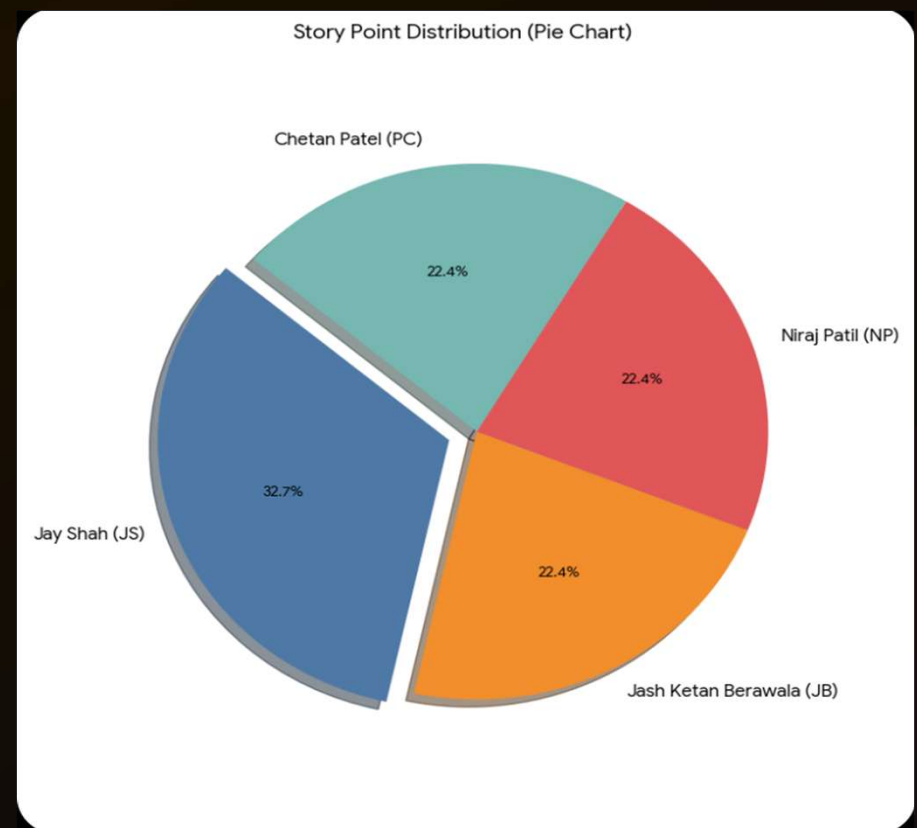


Sprint Velocity Metrics

A team's velocity chart tracks the amount of work completed in each sprint over a series of sprints, enabling the identification of trends and the prediction of future capacity. Velocity typically reflects the story points or work units a team can realistically commit to and successfully complete in a given iteration, a concept further explained by .

Here is the velocity data for "Capstone Sprint 2":

- **Sprint Name:** Capstone Sprint 2
- **Total Committed Points:** 49
- **Total Completed Points:** 49
- Zero Spillover
- **Velocity:** 49
- Team Member Velocity Chart



Sprint burndown



49 points done, 0 points to go



Technical Cards

 Add epic /

 FF-24

Frontend Development – Landing Page and Event Catalog

Done ▾

✓ Done



⌘ Improve Story

▽ Key details

Priority

 Medium

 Add epic /

 FF-25

Implement Login Authentication API using Node.js

Done ▾

✓ Done



⌘ Improve Story

▽ Key details

Priority

 Medium

Deliverables 2

All work

Apps ▾Share ▾

BasicJQLQ Search workSpace ▾Assignee ▾Type ▾Status ▾Key ×More filters ▾Clear filtersSave filter

☐	Work	Assignee	Reporter	Priority	↑ ...	Status
☐	☒ FF-41 Sprint 3 Deliverables	PC Patel Chetan	PC Patel Chetan	= Medium		DONE ▾
☐	FF-36 Selfie scanning by guest	JS jay shah	PC Patel Chetan	= Medium		DONE ▾
☐	FF-35 Basic Image Upload	JS jay shah	PC Patel Chetan	= Medium		DONE ▾
☐	FF-34 Connecting to database/storage	NP Niraj Patil	PC Patel Chetan	= Medium		DONE ▾
☐	FF-33 Guest Landing page	NP Niraj Patil	PC Patel Chetan	^ High		DONE ▾
☐	FF-32 Unique QR code generation for event	PC Patel Chetan	PC Patel Chetan	^ High		DONE ▾
☐	FF-31 Backend photo filtration(Face detection pipeline, ex...	JB Jash Ketan Berawala	PC Patel Chetan	= Medium		DONE ▾
☐	FF-30 Basic Image Upload by the photographer for an ev...	JB Jash Ketan Berawala	PC Patel Chetan	^ High		DONE ▾

8 of 8 ↺

Deliverables 2 Velocity Metrics.

49

Committed

story points

49

Completed

story points

0

Spillover

story points

49

Velocity

pts / sprint

Sprint Velocity — Story Points Completed



Deliverables 3

☐	Work	Assignee	Reporter	Priority	↑ ...	Status
☐	☒ FF-41 Sprint 3 Deliverables	 Patel Chetan	 Patel Chetan	= Medium		DONE 
☐	 FF-36 Selfie scanning by guest	 jay shah	 Patel Chetan	= Medium		DONE 
☐	 FF-35 Basic Image Upload	 jay shah	 Patel Chetan	= Medium		DONE 
☐	 FF-34 Connecting to database/storage	 Niraj Patil	 Patel Chetan	= Medium		DONE 
☐	 FF-33 Guest Landing page	 Niraj Patil	 Patel Chetan	^ High		DONE 
☐	 FF-32 Unique QR code generation for event	 Patel Chetan	 Patel Chetan	^ High		DONE 
☐	 FF-31 Backend photo filtration(Face detection pipeline, ex...	 Jash Ketan Berawala	 Patel Chetan	= Medium		DONE 
☐	 FF-30 Basic Image Upload by the photographer for an ev...	 Jash Ketan Berawala	 Patel Chetan	^ High		DONE 

SPRINT 3

Deliverables 3 Velocity Metrics.

61

Committed

story points

64

Completed

story points

0

Spillover

story points

57.5

Avg Velocity

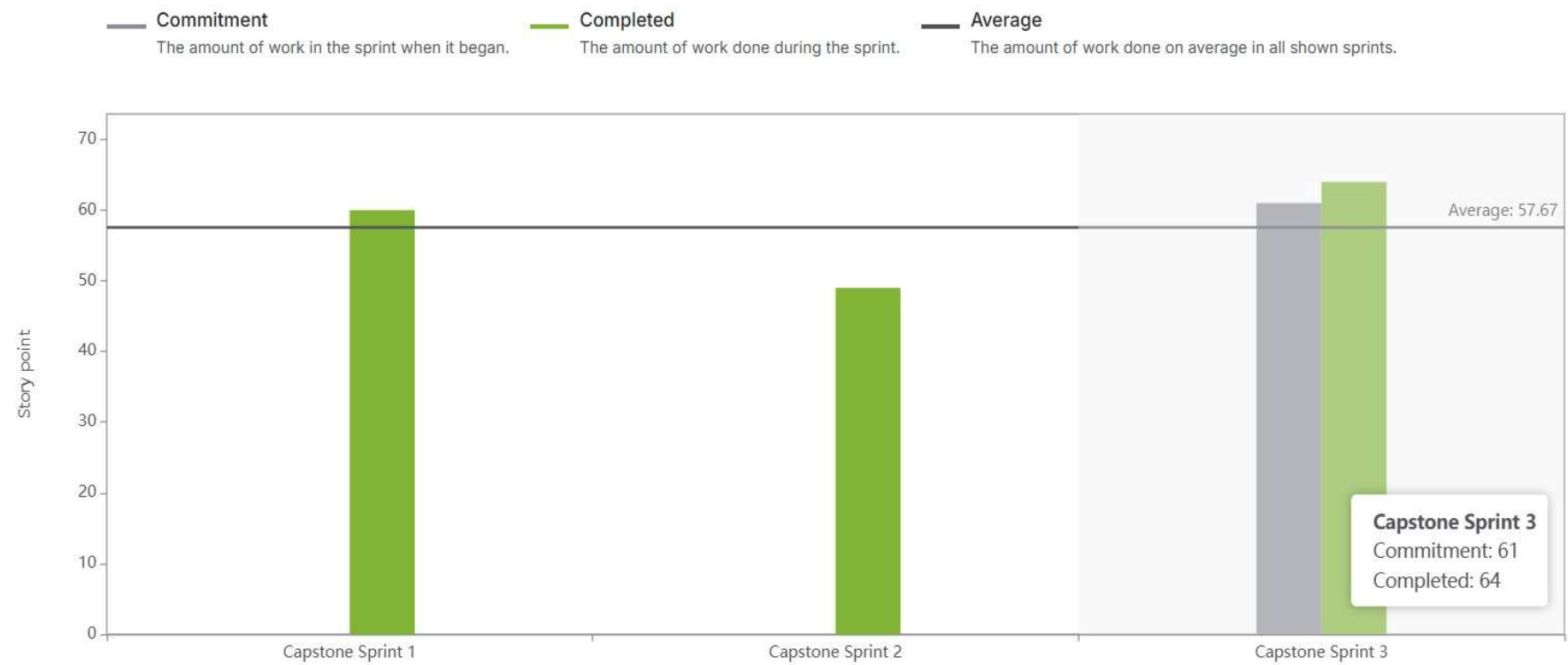
pts / sprint

Team Velocity Trend — All 3 Sprints



Velocity report

[How to read this report](#)

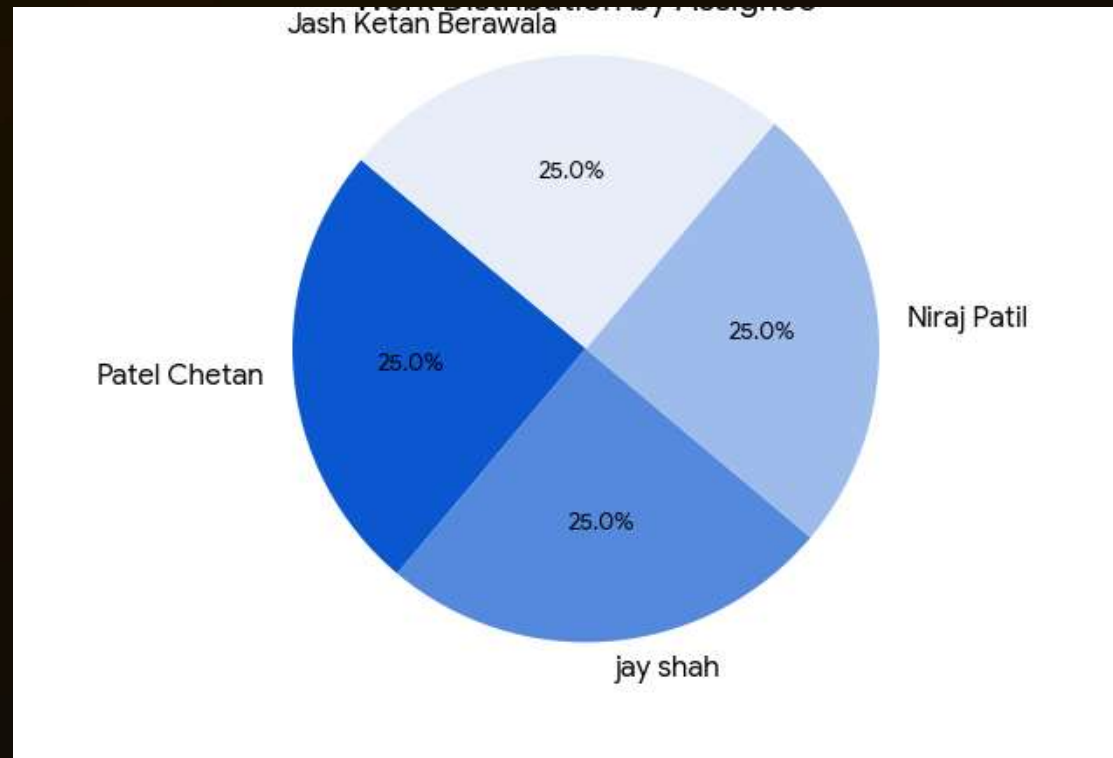


Sprint Velocity Metrics

A team's velocity chart tracks the amount of work completed in each sprint over a series of sprints, enabling the identification of trends and the prediction of future capacity. Velocity typically reflects the story points or work units a team can realistically commit to and successfully complete in a given iteration, a concept further explained by .

Here is the velocity data for "Capstone Sprint 3":

- **Sprint Name:** Capstone Sprint 3
- **Total Committed Points:** 61
- **Total Completed Points:** 64
- Zero Spillover
- **Average Velocity:** 57.5
- Team Member Velocity Chart



TESTING

Test Cases — 53 Cases Across 9 Modules.

Authentication

7

7 test cases

Event Management

8

8 test cases

Photo Upload

7

7 test cases

Face Recognition

6

6 test cases

Guest Portal

6

6 test cases

QR Code System

5

5 test cases

WebSocket/RT

5

5 test cases

MongoDB Storage

4

4 test cases

Admin Dashboard

5

5 test cases

Total: 53 Test Cases · All Passed · 0 Failures

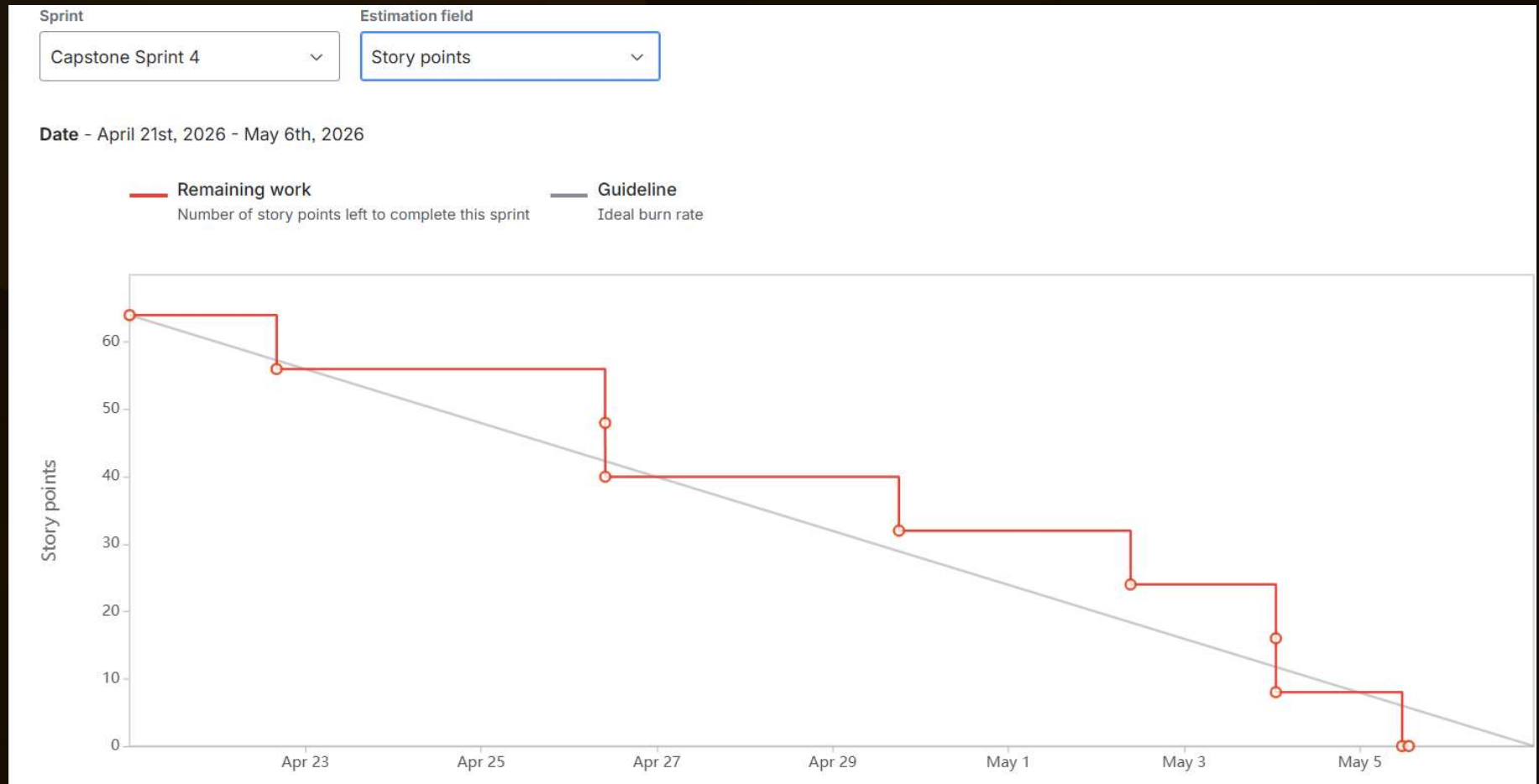
FaceFind · CS 691 · Team 7

Deliverables 4

Deliverables 4 (Completed Tasks)

☐ ▾ Capstone Sprint 4 21 Apr – 6 May (8 work items)		0	0	64	Complete sprint	...
☐ FF-29	Selfie Processing Logic	DONE ▾	8	=	PC	
☐ FF-37	Similarity Matching Engine	DONE ▾	8	=	JB	
☐ FF-38	Personalized Results Gallery	DONE ▾	8	=	JB	
☐ FF-39	Download & Sharing	DONE ▾	8	=	NP	
☒ FF-44	API Documentation	DONE ▾	8		NP	
☒ FF-43	Deployment Guide	DONE ▾	8		JS	
☒ FF-42	Sprint 4 Deliverables	DONE ▾	8		PC	
☒ FF-45	Installation Manual	DONE ▾	8		JS	

Burnup Report



SPRINT 4

Deliverables 4 Velocity Metrics.

64

Committed

story points

64

Completed

story points

0

Spillover

story points

59.25

Avg Velocity

pts / sprint

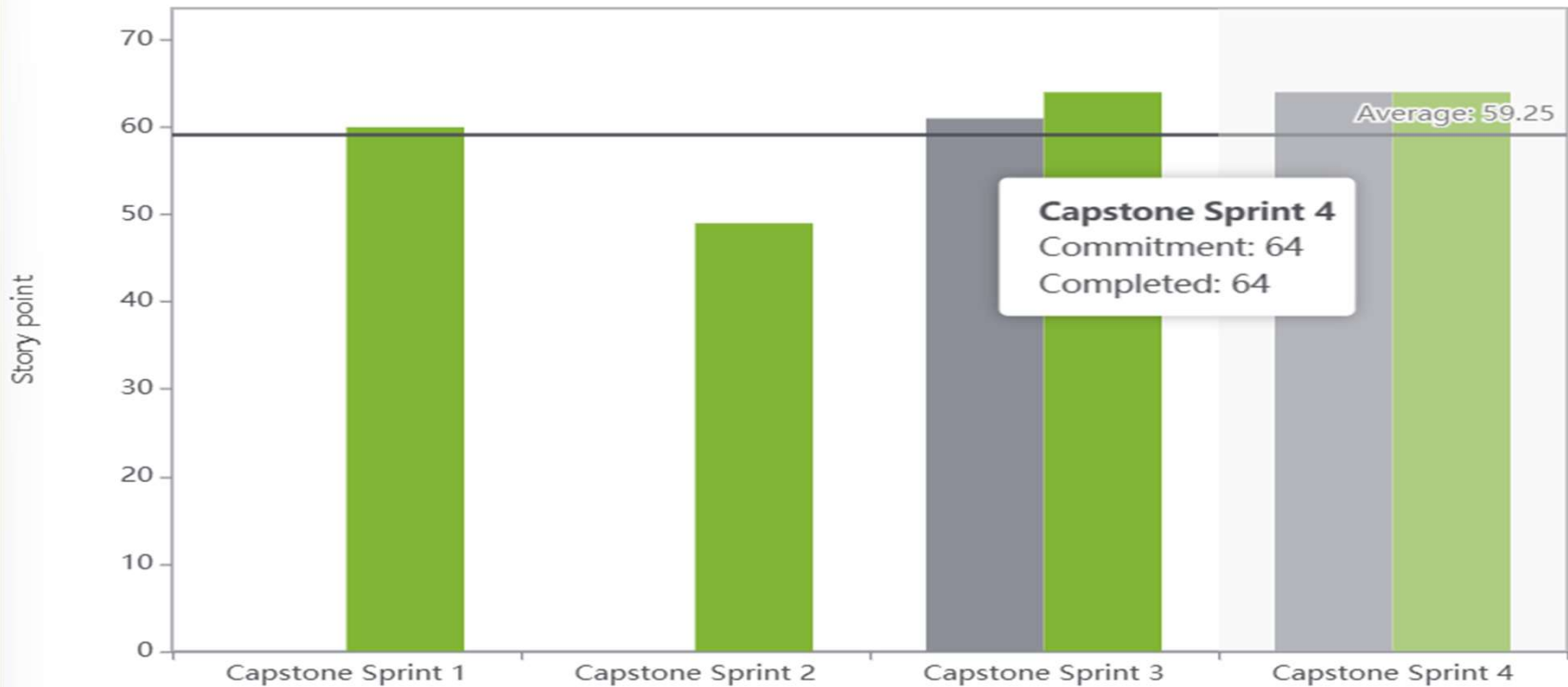
Team Velocity Trend — All 4 Sprints



Velocity report

> How to read this report

- Commitment**
The amount of work in the sprint when it began.
- Completed**
The amount of work done during the sprint.
- Average**
The amount of work done on average in all shown sprints.

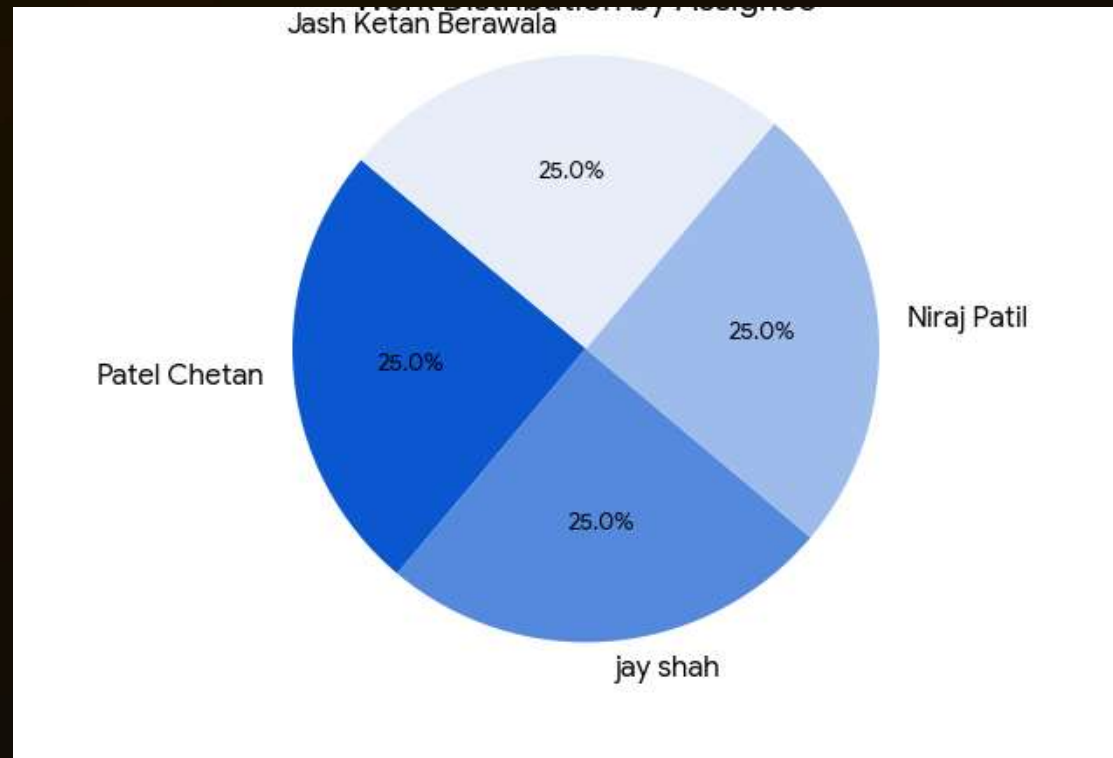


Sprint Velocity Metrics

A team's velocity chart tracks the amount of work completed in each sprint over a series of sprints, enabling the identification of trends and the prediction of future capacity. Velocity typically reflects the story points or work units a team can realistically commit to and successfully complete in a given iteration, a concept further explained by .

Here is the velocity data for "Capstone Sprint 4":

- **Sprint Name:** Capstone Sprint 4
- **Total Committed Points:** 64
- **Total Completed Points:** 64
- Zero Spillover
- **Average Velocity:** 59.25
- Team Member Velocity Chart



Test Cases

FaceFind — AI-Powered Event Photography · Test Case Repository

Application Status: 80% Developed | Phase: Integration & Acceptance Testing | Version: 1.0

Test ID	Module / Feature	Unit to Test	Assumptions	Test Data	Steps to Execute	Expected Result
► AUTHENTICATION TESTS						
TC-AUTH-01	Authentication	Photographer Registration	System is live and accessible	Name: Alex Reynolds Email: alex@moments.com Password: SecurePass@123	1. Navigate to /register 2. Fill all fields 3. Click 'Create Account'	Account created. Confirmation email sent. Redirect to dashboard.
TC-AUTH-02	Authentication	Photographer Login - Valid	Account exists in MongoDB	Email: alex@moments.com Password: SecurePass@123	1. Go to /login 2. Enter valid credentials 3. Click Login	JWT token issued. User redirected to photographer dashboard.
TC-AUTH-03	Authentication	Photographer Login - Invalid Password	Account exists	Email: alex@moments.com Password: wrongpass	1. Go to /login 2. Enter wrong password 3. Click Login	Error: 'Invalid credentials'. No token issued.
TC-AUTH-04	Authentication	Login - Empty Fields	Login form loaded	Email: (empty) Password: (empty)	1. Submit login form with empty fields	Validation error: 'Username and Password are required'
TC-AUTH-05	Authentication	JWT Token Expiry	User is logged in	Expired JWT token	1. Wait for token to expire 2. Attempt to access dashboard	Session expired message. Redirect to /login.
► EVENT MANAGEMENT TESTS						
TC-AUTH-06	Authentication	Duplicate Email Registration	Email already in system	Email: alex@moments.com (existing)	1. Attempt to register with existing email	Error: 'Email already in use'.
TC-EVT-01	Event Management	Create New Event	Photographer logged in	Event Name: Summer Wedding 2025 Date: 2025-06-15 Location: Denver, CO	1. Click 'Create Event' 2. Fill event details 3. Submit	Event created in MongoDB. QR code auto-generated. Event visible in dashboard.
TC-EVT-02	Event Management	View All Events	3+ events in DB for photographer	Photographer: alex@moments.com	1. Login 2. Navigate to My Events	All events listed with name, date, photo count, and QR code link.

FaceFind — AI-Powered Event Photography · Test Case Repository

Application Status: 80% Developed | Phase: Integration & Acceptance Testing | Version: 1.0

Test ID	Module / Feature	Unit to Test	Assumptions	Test Data	Steps to Execute	Expected Result
▶ PHOTO UPLOAD TESTS						
TC-EVT-04	Event Management	Delete Event	Event exists with uploaded photos	Event ID: EVT_001	1. Click delete on EVT_001 2. Confirm deletion	Event, photos, and face embeddings removed from S3 and MongoDB.
TC-EVT-05	Event Management	QR Code Generation	Event successfully created	Event ID: EVT_002	1. Create event 2. View event details	Unique QR code generated per event, links to guest portal.
TC-PHO-01	Photo Upload	Bulk Photo Upload – Valid Files	AWS S3 configured, event created	50 JPG files, avg 4MB each	1. Select event 2. Upload 50 photos 3. Confirm upload	All photos stored in S3. Face processing triggered via FastAPI. Real-time progress shown via WebSocket.
TC-PHO-02	Photo Upload	Upload Invalid File Type	Event exists	File: document.pdf	1. Select event 2. Attempt to upload PDF	Error: 'Only JPG, JPEG, PNG files are accepted'.
TC-PHO-03	Photo Upload	Upload Oversized File	Event exists	File: 50MB RAW image	1. Attempt to upload 50MB file	Error: 'File exceeds maximum size limit'. Upload rejected.

FaceFind — AI-Powered Event Photography · Test Case Repository

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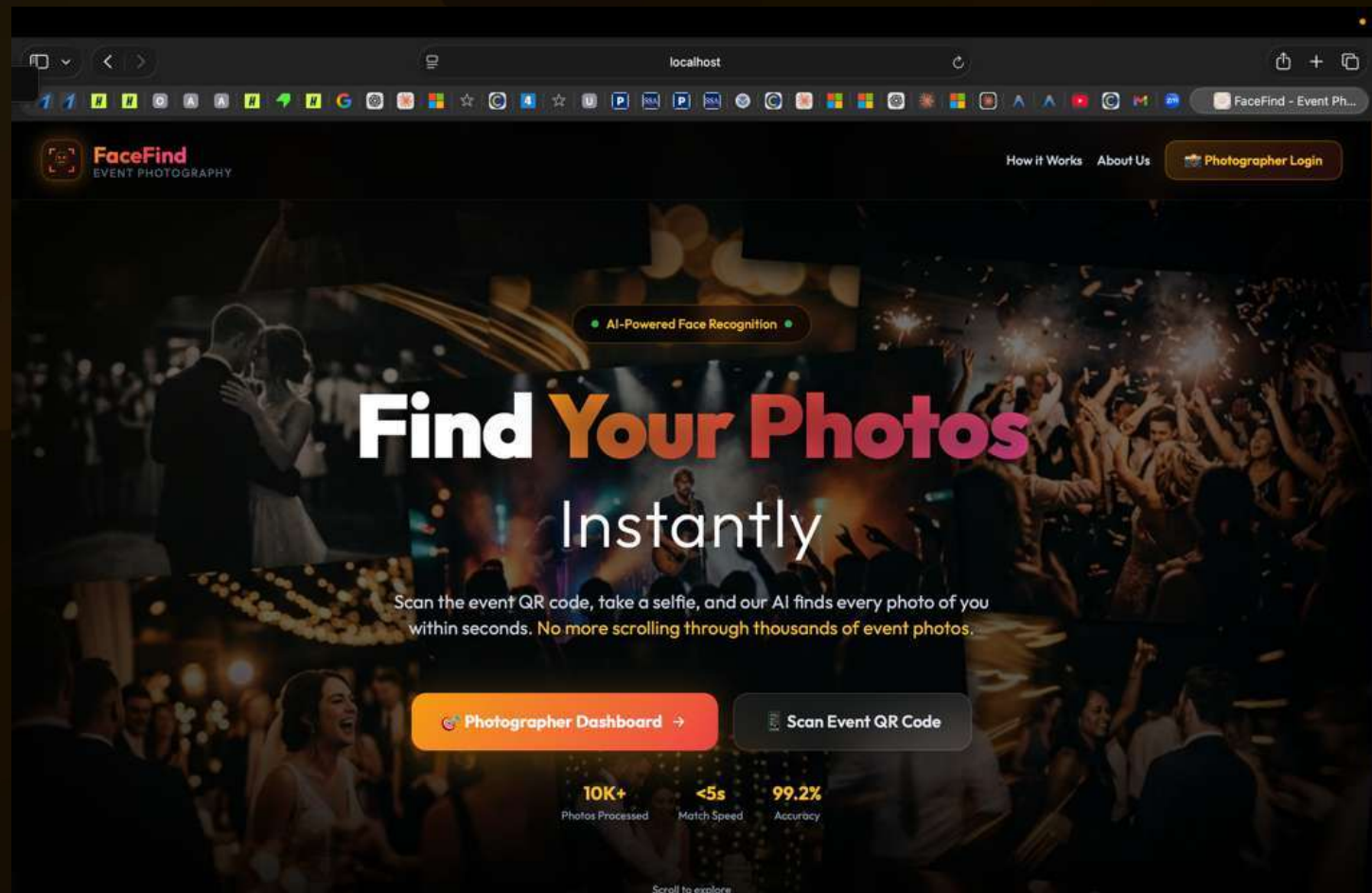
Test ID	Module / Feature	Unit to Test	Assumptions	Test Data	Steps to Execute	Expected Result
TC-PHD-03	Photo Upload	Upload Oversized File	Event exists	File: 50MB RAW image	1. Attempt to upload 50MB file	Error: 'File exceeds maximum size limit'. Upload rejected.
▶ GUEST PORTAL TESTS						
TC-PHD-04	Photo Upload	Real-time Processing Dashboard	WebSocket connection active	30 photos uploaded	1. Upload batch of photos 2. Observe dashboard	Dashboard shows live count of processed photos. Percentage updates in real time.
TC-PHD-05	Photo Upload	AI Face Indexing on Upload	ML microservice running (FastAPI)	10 photos with clear faces	1. Upload photos 2. Wait for processing	Face embeddings generated and stored in vector DB. Each face indexed with photo reference.
TC-PHD-06	Photo Upload	Upload Photo with No Faces	Event exists, ML service active	Landscape photo (no people)	1. Upload landscape photo	Photo stored in S3. No face embedding generated. No error thrown.
TC-GST-01	Guest Portal	QR Code Scan Access	Event is active, QR code valid	Mobile device with camera	1. Scan event QR code 2. Observe redirect	Browser opens guest portal. No app download required.
TC-GST-02	Guest Portal	Selfie Upload – Valid Face	Guest portal loaded, ML running	Clear front-facing selfie JPG	1. Scan QR 2. Upload selfie 3. Submit	Face matched against event photos. Results returned in < 5 seconds.
TC-GST-03	Guest Portal	Selfie Upload – No Face Detected	Guest portal active	Selfie: image of a wall	1. Upload non-face image as selfie	Error: 'No face detected. Please upload a clear selfie'.

Test ID	Module / Feature	Unit to Test	Assumptions	Test Data	Steps to Execute	Expected Result
		Detected				
► FACE RECOGNITION / ML TESTS						
TC-GST-04	Guest Portal	Face Match Returns Results	ML service active, 100 photos indexed	Selfie of person present in 8 event photos	1. Upload selfie 2. Wait for match	All 8 matching photos displayed in gallery. Match accuracy verified.
TC-GST-05	Guest Portal	No Match Found	Event photos indexed	Selfie of person not in event	1. Upload selfie of non-attendee	Message: 'No photos found matching your face'. No results shown.
TC-GST-06	Guest Portal	Selfie Deleted After Matching	Privacy policy enforced in backend	Valid selfie uploaded	1. Upload selfie 2. Receive results 3. Check S3 for selfie	Selfie not retained in S3 after matching process completes.
TC-GST-07	Guest Portal	Expired QR Code / Inactive Event	Event set to expired	QR code from expired event	1. Scan expired QR code	Error: 'This event has ended or is no longer active'.
► CORPORATE EVENT TESTS						
TC-ML-01	Face Recognition	DBSCAN Face Clustering	Python FastAPI ML service running	50 photos, 10 unique individuals	1. Upload 50 photos 2. Trigger clustering	Faces clustered into ~10 groups using DBSCAN. No significant misclassification.
TC-ML-02	Face Recognition	Match Speed < 5 Seconds	100 photos indexed	Selfie of indexed person	1. Upload selfie 2. Start timer 3. Receive results	Results returned in < 5 seconds (SLA requirement).

FINAL PROJECT DEMO



First Page (Demo)



Login page

FaceFind
STUDIO

★★★★★

"FaceFind transformed how I deliver photos to my clients. They find their shots instantly — it's like magic."

SP Sarah Parker
Wedding Photographer, NYC

500+
Photographers

10K+
Events

2M+
Photos Matched

Sign In **Sign Up**

Welcome back
Sign in to your photographer dashboard

Email
u@studio.com

Password
Enter your password

☐ Remember me [Forgot password?](#)

Sign in to Studio

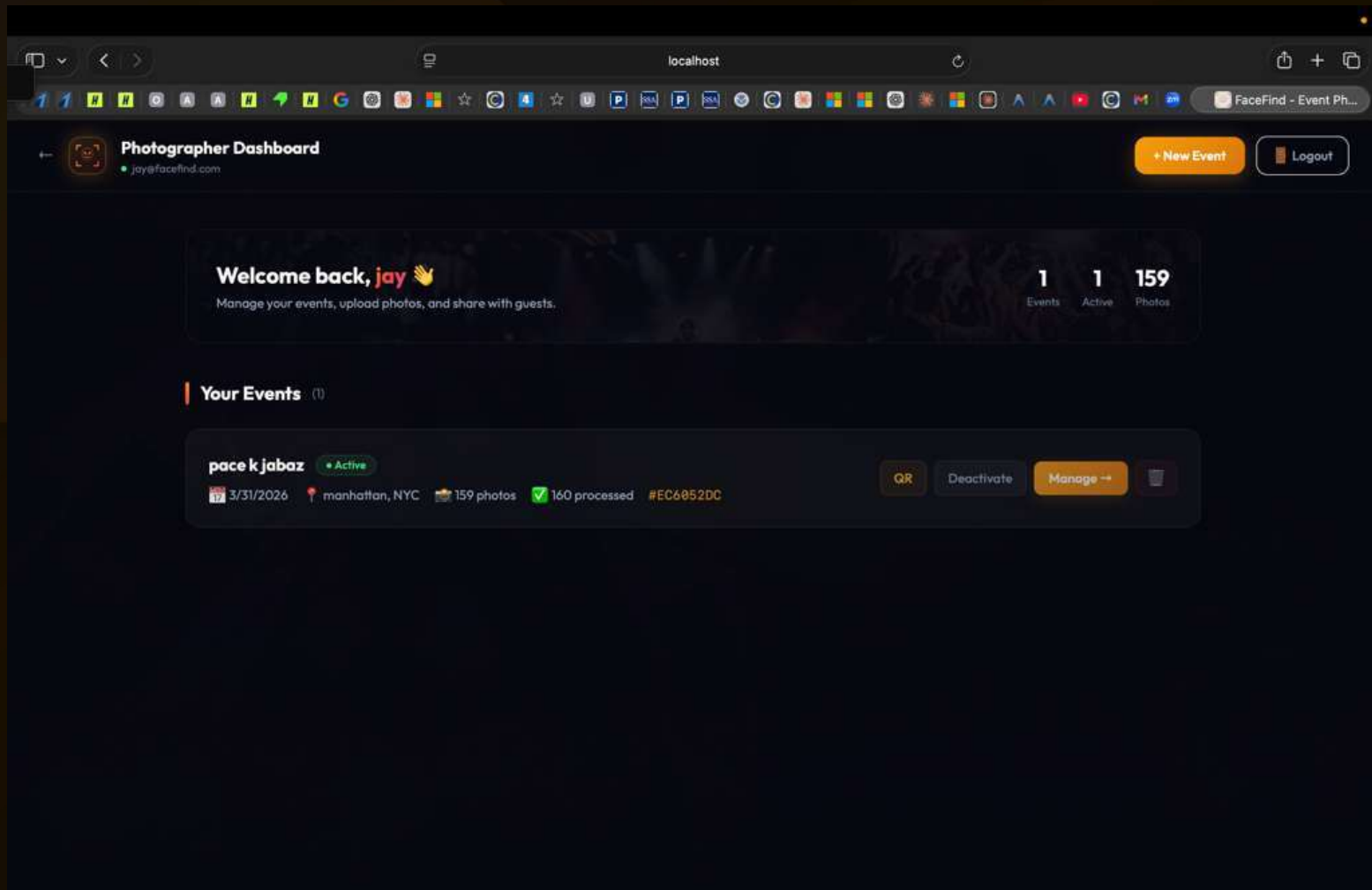
OR

Demo Mode: any email & password
[Use Demo Credentials](#)

Don't have an account? [Sign Up](#)

[Back to Home](#)

Dashboard Page (after Login)



CREATE AN EVENT POP-UP

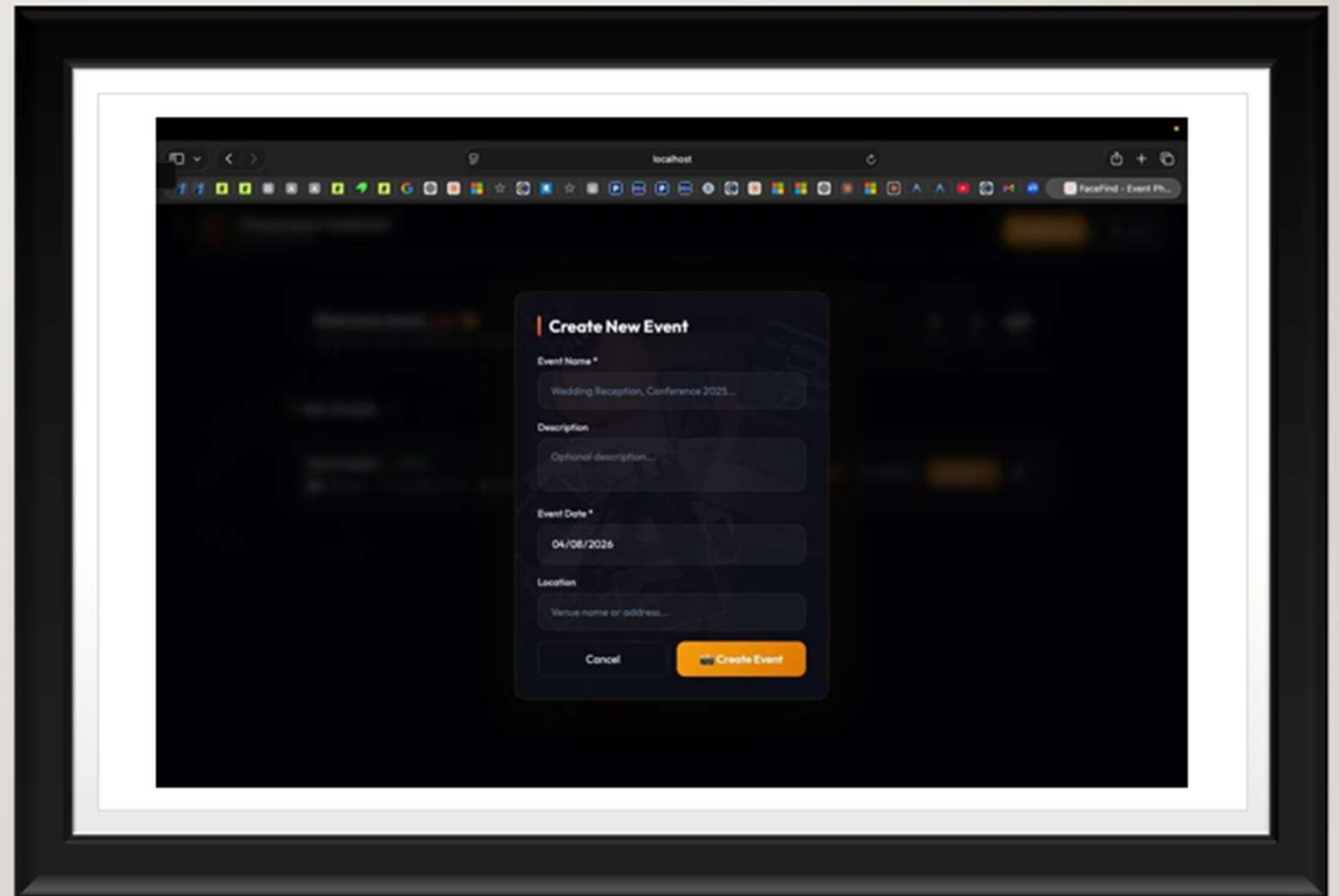
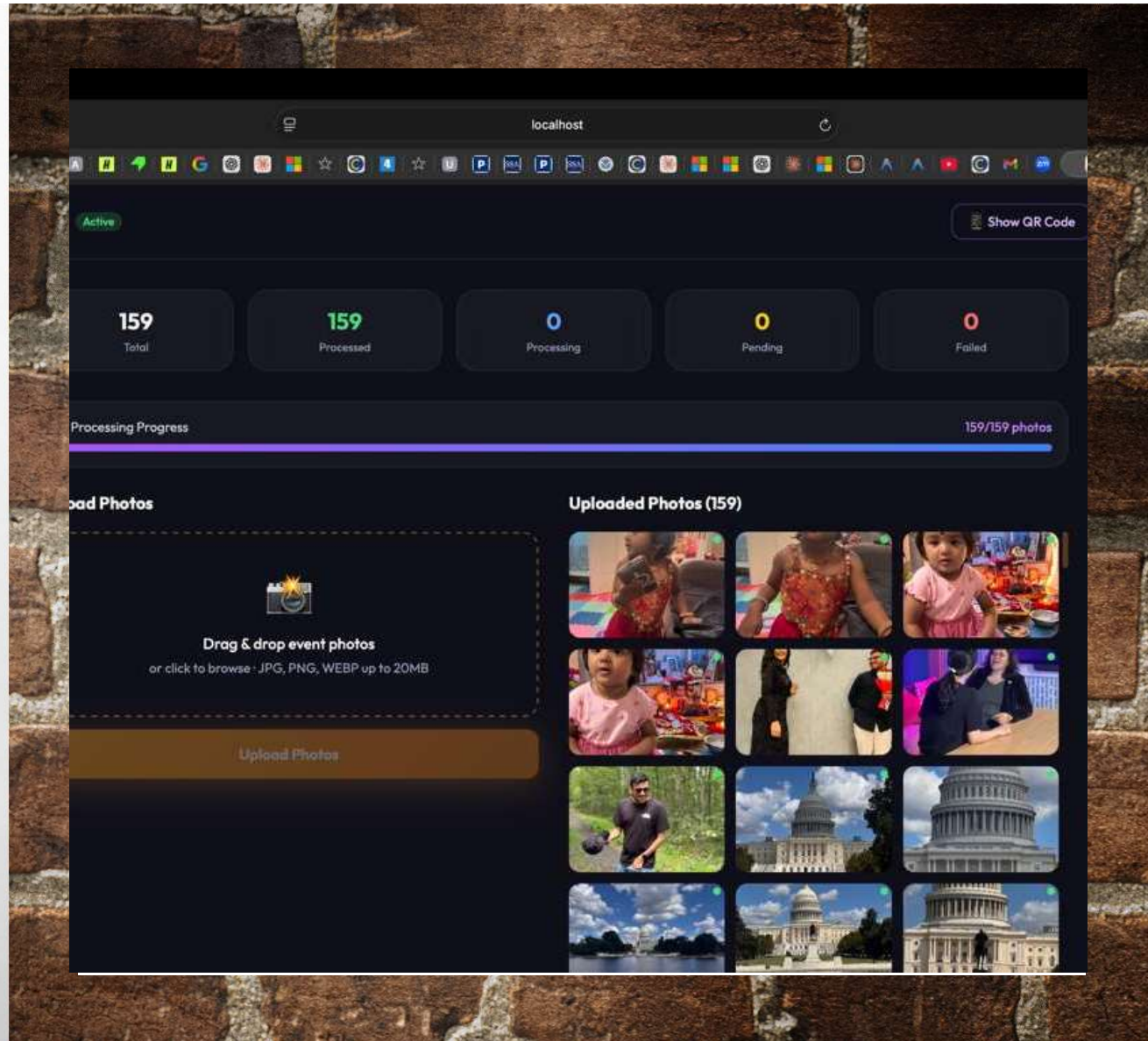
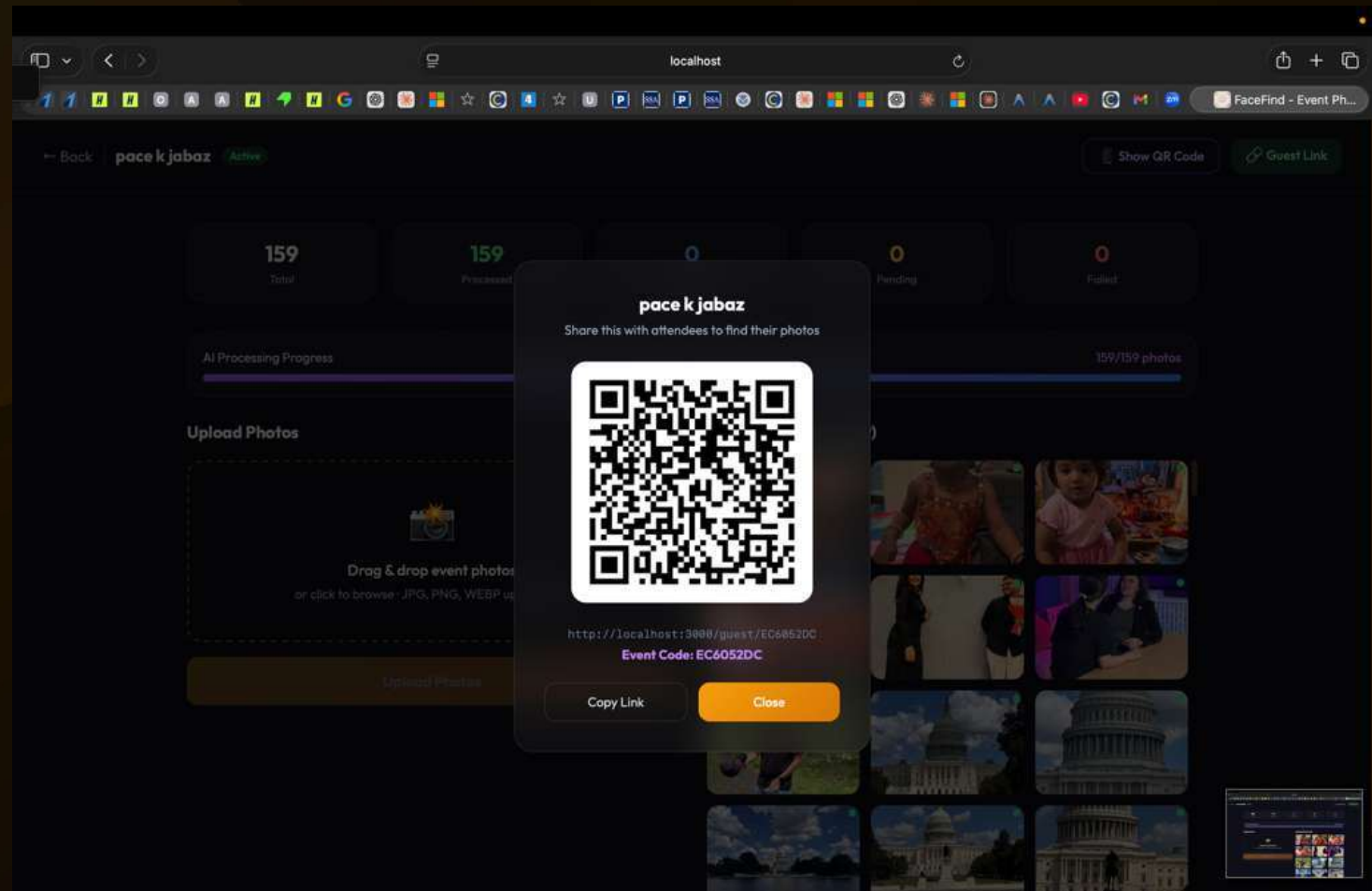


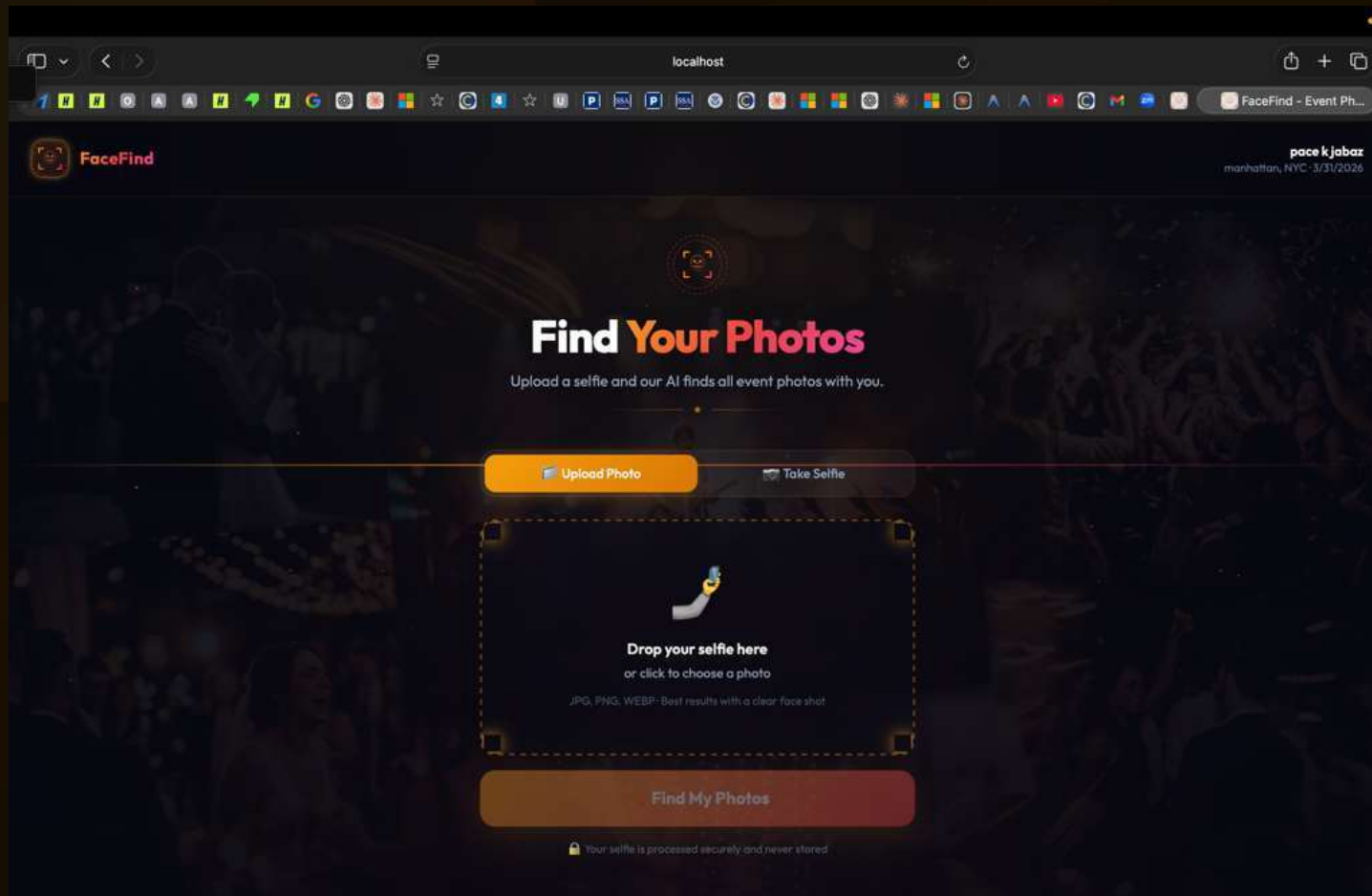
IMAGE UPLOAD DASHBOARD(FOR PHOTOGRAPHER)



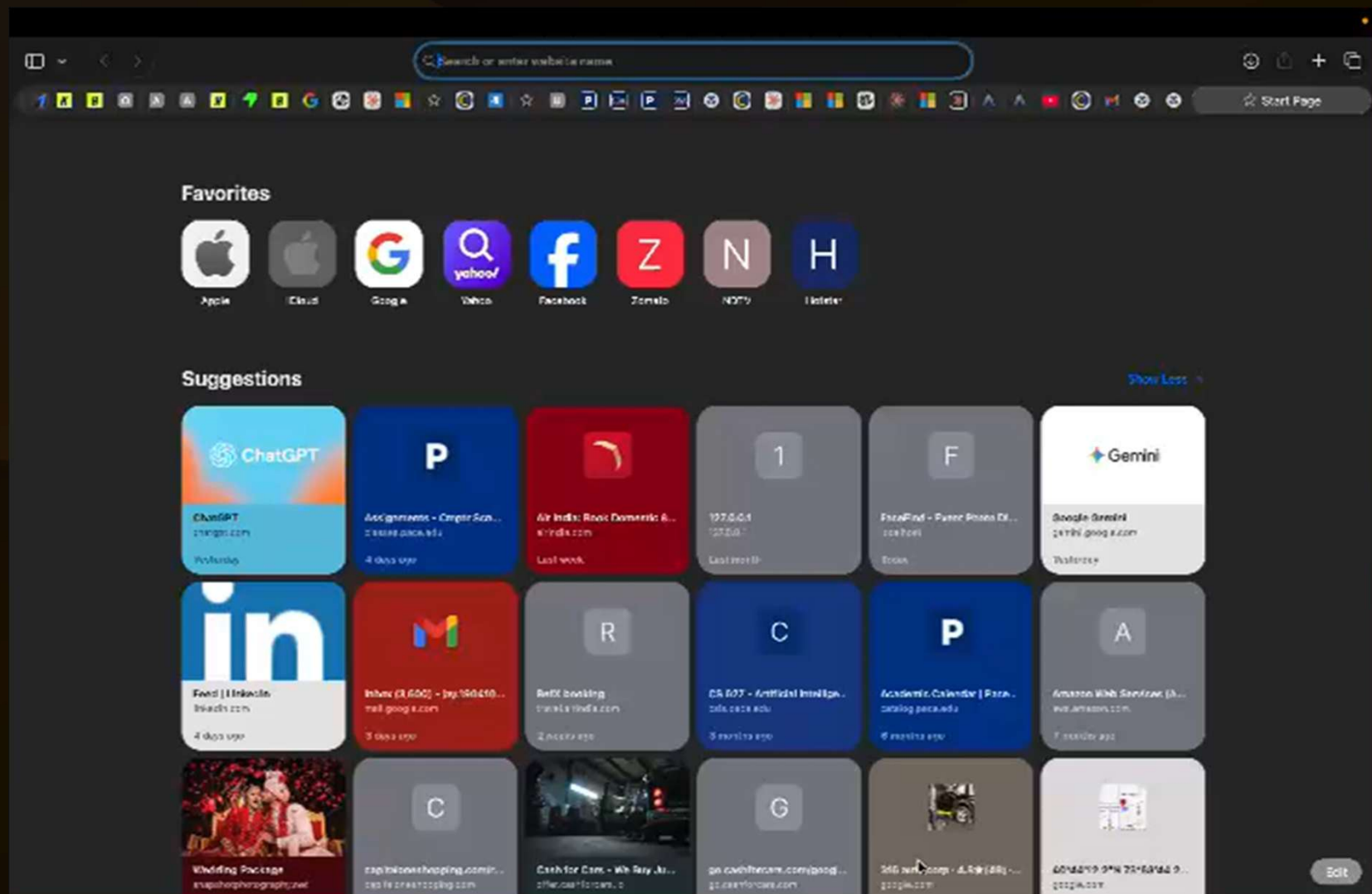
Guest QR Code Scanning Link



Guest Page To find photos after scanning QR Code



PROJECT DEMO VIDEO



Deliverables 4 Retrospective.

✓ WENT WELL

- 0 spillover & 100% tasks achieved
- Core features shipped end-to-end
- Strong team collaboration & task distribution
- Improved Jira ticket hygiene
- Unit testing by developer and, also tested by other teammates

⚠ IMPROVE

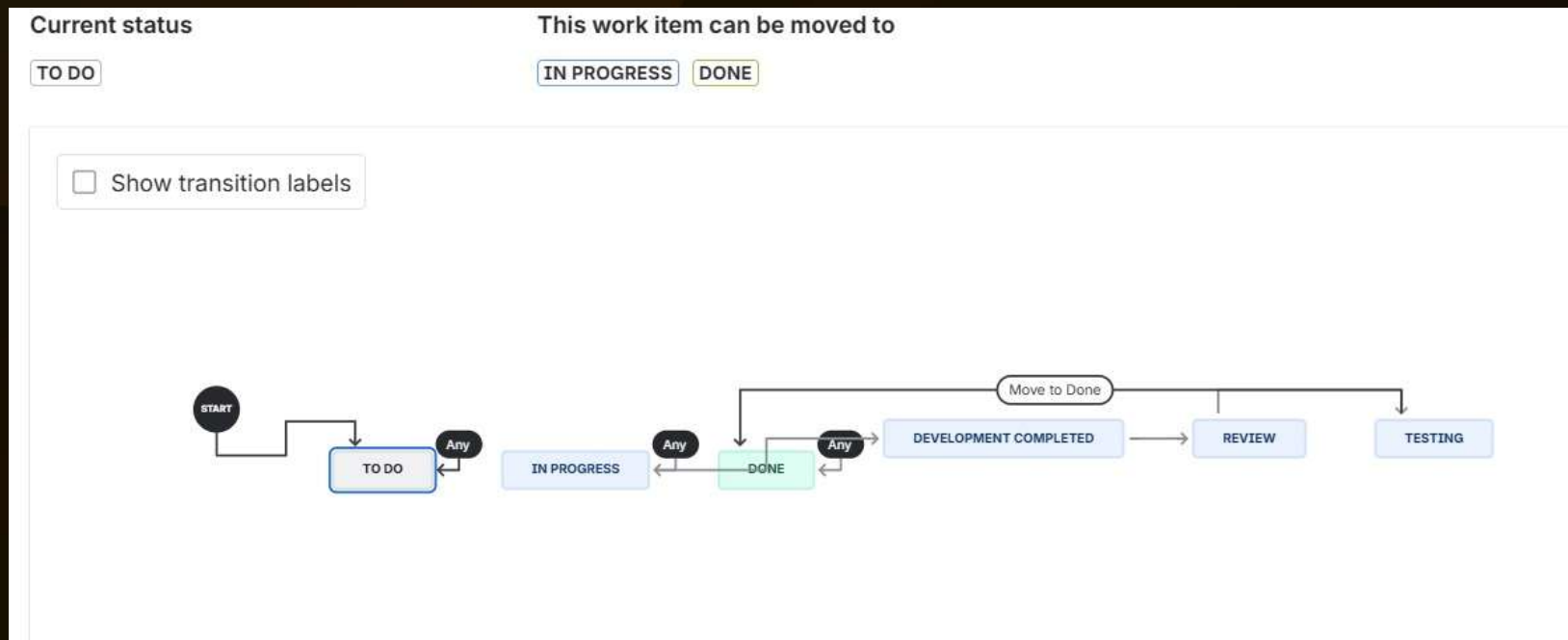
- Add QA/testing tickets to Jira
- Combined Regression testing card to be created for all technical cards in each sprint

🎯 ACTION ITEMS

- Add dedicated QA tasks in Sprint 4
- Continue 3-day deep-dive session format

Jira Workflow

- Each ticket was assigned to other person for testing apart from the developer so that the technician feature can be tested twice (Unit testing by developer and, also tested by other teammates)



Overall Summary

- After identifying the problem successfully implemented the solution and created the working application.
- Completed all the sprints with zero spillover.
- Completed the project with Scrum guidelines and successfully delivered it.

FF

Thank You!

Questions Welcome.



Group Wiki

github.com/JashBerawala/Pace-Project-Team7/wiki



GitHub Repo

github.com/JashBerawala/Pace-Project-Team7



Team 7

CS 691 · Pace University